Neural Computation -- 2010

102 article(s), all issues

1. Growing self-organizing surface map

MoleVilson Luiz Dalle, AraújoAluizio Fausto Ribeiro

(no DOI)

The growing self-organizing surface map (GSOSM) is a novel map model that learns a folded surface immersed in a 3D space. Starting from a dense point cloud, the surface is reconstructed through an ...

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2. Rademacher chaos complexities for learning the kernel problem

YingYiming, CampbellColin

(no DOI)

We develop a novel generalization bound for learning the kernel problem. First, we show that the generalization analysis of the kernel learning problem reduces to investigation of the suprema of th...

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3. Model-Based Decoding, Information Estimation, and Change-Point Detection Techniques for Multineuron Spike Trains

Jonathan W. Pillow, Yashar Ahmadian, Liam Paninski

https://doi.org/10.1162/neco_a_00058 | pp. 1-45

One of the central problems in systems neuroscience is to understand how neural spike trains convey sensory information. Decoding methods, which provide an explicit means for reading out the information contained in neural spike responses, offer a powerful set of tools for studying the neural coding problem. Here we develop several decoding methods based on point-process neural encoding models, or forward models that predict spike responses to stimuli. These models have concave log-likelihood functions, which allow efficient maximum-likelihood model fitting and stimulus decoding. We present several applications of the encoding model framework to the problem of decoding stimulus information from population spike responses: (1) a tractable algorithm for computing the maximum a posteriori (MAP) estimate of the stimulus, the most probable stimulus to have generated an observed single- or multiple-neuron spike train response, given some prior distribution over the stimulus; (2) a gaussian approximation to the posterior stimulus distribution that can be used to quantify the fidelity with which various stimulus features are encoded; (3) an efficient method for estimating the mutual information between the stimulus and the spike trains emitted by a neural population; and (4) a framework for the detection of change-point times (the time at which the stimulus undergoes a change in mean or variance) by marginalizing over the posterior stimulus distribution. We provide several examples illustrating the performance of these estimators with simulated and real neural data.

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4. Efficient Markov Chain Monte Carlo Methods for Decoding Neural Spike Trains

Yashar Ahmadian, Jonathan W. Pillow, Liam Paninski
https://doi.org/10.1162/neco_a_00059 | pp. 46-96

Stimulus reconstruction or decoding methods provide an important tool for understanding how sensory and motor information is represented in neural activity. We discuss Bayesian decoding methods based on an encoding generalized linear model (GLM) that accurately describes how stimuli are transformed into the spike trains of a group of neurons. The form of the GLM likelihood ensures that the posterior distribution over the stimuli that caused an observed set of spike trains is log concave so long as the prior is. This allows the maximum a posteriori (MAP) stimulus estimate to be obtained using efficient optimization algorithms. Unfortunately, the MAP estimate can have a relatively large average error when the posterior is highly nongaussian. Here we compare several Markov chain Monte Carlo (MCMC) algorithms that allow for the calculation of general Bayesian estimators involving posterior expectations (conditional on model parameters). An efficient version of the hybrid Monte Carlo (HMC) algorithm was significantly superior to other MCMC methods for gaussian priors. When the prior distribution has sharp edges and corners, on the other hand, the "hit-and-run" algorithm performed better than other MCMC methods. Using these algorithms, we show that for this latter class of priors, the posterior mean estimate can have a considerably lower average error than MAP, whereas for gaussian priors, the two estimators have roughly equal efficiency. We also address the application of MCMC methods for extracting nonmarginal properties of the posterior distribution. For example, by using MCMC to calculate the mutual information between the stimulus and response, we verify the validity of a computationally efficient Laplace approximation to this quantity for gaussian priors in a wide range of model parameters; this makes direct model-based computation of the mutual information tractable even in the case of large observed neural populations, where methods based on binning the spike train fail. Finally, we consider the effect of uncertainty in the GLM parameters on the posterior estimators.

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5. Modeling Multivariate Time Series on Manifolds with Skew Radial Basis Functions

Arta A. Jamshidi, Michael Kirby
https://doi.org/10.1162/neco_a_00060 | pp. 97-123

We present an approach for constructing nonlinear empirical mappings from high-dimensional domains to multivariate ranges. We employ radial basis functions and skew radial basis functions for constructing a model using data that are potentially scattered or sparse. The algorithm progresses iteratively, adding a new function at each step to refine the model. The placement of the functions is driven by a statistical hypothesis test that accounts for correlation in the multivariate range variables. The test is applied on training and validation data and reveals nonstatistical or geometric structure when it fails. At each step, the added function is fit to data contained in a spatiotemporally defined local region to determine the parameters--in particular, the scale of the local model. The scale of the function is determined by the zero crossings of the autocorrelation function of the residuals. The model parameters and the number of basis functions are determined automatically from the given data, and there is no need to initialize any ad hoc parameters save for the selection of the skew radial basis functions. Compactly supported skew radial basis functions are employed to improve model accuracy, order, and convergence properties. The extension of the algorithm to higher-dimensional ranges produces reduced-order models by exploiting the existence of correlation in the range variable data. Structure is tested not just in a single time series but between all pairs of time series.

We illustrate the new methodologies using several illustrative problems, including modeling data on manifolds and the prediction of chaotic time series.

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6. The Mean Time to Express Synaptic Plasticity in Integrate-and-Express, Stochastic Models of Synaptic Plasticity Induction

Terry Elliott

https://doi.org/10.1162/neco_a_00061 | pp. 124-159

Stochastic models of synaptic plasticity propose that single synapses perform a directed random walk of fixed step sizes in synaptic strength, thereby embracing the view that the mechanisms of synaptic plasticity constitute a stochastic dynamical system. However, fluctuations in synaptic strength present a formidable challenge to such an approach. We have previously proposed that single synapses must interpose an integration and filtering mechanism between the induction of synaptic plasticity and the expression of synaptic plasticity in order to control fluctuations. We analyze a class of three such mechanisms in the presence of possibly non-Markovian plasticity induction processes, deriving expressions for the mean expression time in these models. One of these filtering mechanisms constitutes a discrete low-pass filter that could be implemented on a small collection of molecules at single synapses, such as CaMKII, and we analyze this discrete filter in some detail. After considering Markov induction processes, we examine our own stochastic model of spike-timing-dependent plasticity, for which the probability density functions of the induction of plasticity steps have previously been derived. We determine the dependence of the mean time to express a plasticity step on pre- and postsynaptic firing rates in this model, and we also consider, numerically, the long-term stability against fluctuations of patterns of neuronal connectivity that typically emerge during neuronal development.

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7. A Graphical Model Framework for Decoding in the Visual ERP-Based BCI Speller

Suzanne Martens, Joris M. Mooij, N. Jeremy Hill, Jason Farquhar, Bernhard Schölkopf

https://doi.org/10.1162/neco_a_00066 | pp. 160-182

We present a graphical model framework for decoding in the visual ERP-based speller system. The proposed framework allows researchers to build generative models from which the decoding rules are obtained in a straightforward manner. We suggest two models for generating brain signals conditioned on the stimulus events. Both models incorporate letter frequency information but assume different dependencies between brain signals and stimulus events. For both models, we derive decoding rules and perform a discriminative training. We show on real visual speller data how decoding performance improves by incorporating letter frequency information and using a more realistic graphical model for the dependencies between the brain signals and the stimulus events. Furthermore, we discuss how the standard approach to decoding can be seen as a special case of the graphical model framework. The letter also gives more insight into the discriminative approach for decoding in the visual speller system.

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8. A Framework for Simulating and Estimating the State and Functional Topology of Complex Dynamic Geometric Networks

Marius Buibas, Gabriel A. Silva

https://doi.org/10.1162/neco_a_00065 | pp. 183-214

We introduce a framework for simulating signal propagation in geometric networks (networks that can be mapped to geometric graphs in some space) and developing algorithms that estimate (i.e., map) the state and functional topology of complex dynamic geometric networks. Within the framework, we define the key features typically present in such networks and of particular relevance to biological cellular neural networks: dynamics, signaling, observation, and control. The framework is particularly well suited for estimating functional connectivity in cellular neural networks from experimentally observable data and has been implemented using graphics processing unit high-performance computing. Computationally, the framework can simulate cellular network signaling close to or faster than real time. We further propose a standard test set of networks to measure performance and compare different mapping algorithms.

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9. A Multiscale Correlation of Wavelet Coefficients Approach to Spike Detection

Chenhui Yang, Byron Olson, Jennie Si

https://doi.org/10.1162/neco_a_00063 | pp. 215-250

Extracellular chronic recordings have been used as important evidence in neuroscientific studies to unveil the fundamental neural network mechanisms in the brain. Spike detection is the first step in the analysis of recorded neural waveforms to decipher useful information and provide useful signals for brain-machine interface applications. The process of spike detection is to extract action potentials from the recordings, which are often compounded with noise from different sources. This study proposes a new detection algorithm that leverages a technique from wavelet-based image edge detection. It utilizes the correlation between wavelet coefficients at different sampling scales to create a robust spike detector. The algorithm has one tuning parameter, which potentially reduces the subjectivity of detection results. Both artificial benchmark data sets and real neural recordings are used to evaluate the detection performance of the proposed algorithm. Compared with other detection algorithms, the proposed method has a comparable or better detection performance. In this letter, we also demonstrate its potential for real-time implementation.

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10. Broken Symmetries in a Location-Invariant Word Recognition Network

Thomas Hannagan, Frédéric Dandurand, Jonathan Grainger

https://doi.org/10.1162/neco_a_00064 | pp. 251-283

We studied the feedforward network proposed by Dandurand et al. (2010), which maps location-specific letter inputs to location-invariant word outputs, probing the hidden layer to determine the nature of the code. Hidden patterns for words were densely distributed, and K-means clustering on single letter patterns produced evidence that the network had formed semi-location-invariant letter representations during training. The possible confound with superseding bigram representations was ruled out, and linear regressions showed that any word pattern was well approximated by a linear combination of its constituent letter patterns. Emulating this code using overlapping holographic representations (Plate, 1995) uncovered a

surprisingly acute and useful correspondence with the network, stemming from a broken symmetry in the connection weight matrix and related to the group-invariance theorem (Minsky & Papert, 1969). These results also explain how the network can reproduce relative and transposition priming effects found in humans.

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11. Least-Squares Independent Component Analysis

Taiji Suzuki, Masashi Sugiyama

https://doi.org/10.1162/neco_a_00062 | pp. 284-301

Accurately evaluating statistical independence among random variables is a key element of independent component analysis (ICA). In this letter, we employ a squared-loss variant of mutual information as an independence measure and give its estimation method. Our basic idea is to estimate the ratio of probability densities directly without going through density estimation, thereby avoiding the difficult task of density estimation. In this density ratio approach, a natural cross-validation procedure is available for hyperparameter selection. Thus, all tuning parameters such as the kernel width or the regularization parameter can be objectively optimized. This is an advantage over recently developed kernel-based independence measures and is a highly useful property in unsupervised learning problems such as ICA. Based on this novel independence measure, we develop an ICA algorithm, named least-squares independent component analysis.

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12. A Theory of Slow Feature Analysis for Transformation-Based Input Signals with an Application to Complex Cells

Henning Sprekeler, Laurenz Wiskott

https://doi.org/10.1162/neco_a_00072 | pp. 303-335

We develop a group-theoretical analysis of slow feature analysis for the case where the input data are generated by applying a set of continuous transformations to static templates. As an application of the theory, we analytically derive nonlinear visual receptive fields and show that their optimal stimuli, as well as the orientation and frequency tuning, are in good agreement with previous simulations of complex cells in primary visual cortex (Berkes and Wiskott, 2005). The theory suggests that side and end stopping can be interpreted as a weak breaking of translation invariance. Direction selectivity is also discussed.

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13. Bayes-Optimal Chemotaxis

Duncan Mortimer, Peter Dayan, Kevin Burrage, Geoffrey J. Goodhill

https://doi.org/10.1162/neco_a_00075 | pp. 336-373

Chemotaxis plays a crucial role in many biological processes, including nervous system development. However, fundamental physical constraints limit the ability of a small sensing device such as a cell or growth cone to detect an external chemical gradient. One of these is the stochastic nature of receptor binding, leading to a constantly fluctuating binding pattern across the cell's array of receptors. This is analogous to the uncertainty in sensory information often encountered by the brain at the systems level. Here we derive analytically the Bayes-optimal strategy for combining information from a spatial array of receptors in both

one and two dimensions to determine gradient direction. We also show how information from more than one receptor species can be optimally integrated, derive the gradient shapes that are optimal for guiding cells or growth cones over the longest possible distances, and illustrate that polarized cell behavior might arise as an adaptation to slowly varying environments. Together our results provide closed-form predictions for variations in chemotactic performance over a wide range of gradient conditions.

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14. Least Squares Estimation Without Priors or Supervision

Martin Raphan, Eero P. Simoncelli

https://doi.org/10.1162/neco_a_00076 | pp. 374-420

Selection of an optimal estimator typically relies on either supervised training samples (pairs of measurements and their associated true values) or a prior probability model for the true values. Here, we consider the problem of obtaining a least squares estimator given a measurement process with known statistics (i.e., a likelihood function) and a set of unsupervised measurements, each arising from a corresponding true value drawn randomly from an unknown distribution. We develop a general expression for a nonparametric empirical Bayes least squares (NEBLS) estimator, which expresses the optimal least squares estimator in terms of the measurement density, with no explicit reference to the unknown (prior) density. We study the conditions under which such estimators exist and derive specific forms for a variety of different measurement processes. We further show that each of these NEBLS estimators may be used to express the mean squared estimation error as an expectation over the measurement density alone, thus generalizing Stein's unbiased risk estimator (SURE), which provides such an expression for the additive gaussian noise case. This error expression may then be optimized over noisy measurement samples, in the absence of supervised training data, yielding a generalized SURE-optimized parametric least squares (SURE2PLS) estimator. In the special case of a linear parameterization (i.e., a sum of nonlinear kernel functions), the objective function is quadratic, and we derive an incremental form for learning this estimator from data. We also show that combining the NEBLS form with its corresponding generalized SURE expression produces a generalization of the score-matching procedure for parametric density estimation. Finally, we have implemented several examples of such estimators, and we show that their performance is comparable to their optimal Bayesian or supervised regression counterparts for moderate to large amounts of data.

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15. Improved Integral Equation Solution for the First Passage Time of Leaky Integrate-and-Fire Neurons

Yi Dong, ■tefan Mihala■, Ernst Niebur

https://doi.org/10.1162/neco_a_00078 | pp. 421-434

An accurate calculation of the first passage time probability density (FPTPD) is essential for computing the likelihood of solutions of the stochastic leaky integrate-and-fire model. The previously proposed numerical calculation of the FPTPD based on the integral equation method discretizes the probability current of the voltage crossing the threshold. While the method is accurate for high noise levels, we show that it results in large numerical errors for small noise. The problem is solved by analytically computing, in each time bin, the mean probability current. Efficiency is further improved by identifying and ignoring time bins with negligible mean probability current.

16. Abstract Stimulus-Specific Adaptation Models

Robert Mill, Martin Coath, Thomas Wennekers, Susan L. Denham

https://doi.org/10.1162/neco_a_00077 | pp. 435-476

Many neurons that initially respond to a stimulus stop responding if the stimulus is presented repeatedly but recover their response if a different stimulus is presented. This phenomenon is referred to as stimulus-specific adaptation (SSA). SSA has been investigated extensively using oddball experiments, which measure the responses of a neuron to sequences of stimuli. Neurons that exhibit SSA respond less vigorously to common stimuli, and the metric typically used to quantify this difference is the SSA index (SI). This article presents the first detailed analysis of the SI metric by examining the question: How should a system (e.g., a neuron) respond to stochastic input if it is to maximize the SI of its output? Questions like this one are particularly relevant to those wishing to construct computational models of SSA. If an artificial neural network receives stimulus information at a particular rate and must respond within a fixed time, what is the highest SI one can reasonably expect? We demonstrate that the optimum, average SI is constrained by the information in the input source, the length and encoding of the memory, and the assumptions concerning how the task is decomposed.

17. Modeling Basal Ganglia for Understanding Parkinsonian Reaching Movements

Kulam Najmudeen Magdoom, Deepak Subramanian, V. Srinivasa Chakravarthy, Balaraman Ravindran, ■■■-■■■ ■■■■■, N. Meenakshisundaram

https://doi.org/10.1162/neco_a_00073 | pp. 477-516

We present a computational model that highlights the role of basal ganglia (BG) in generating simple reaching movements. The model is cast within the reinforcement learning (RL) framework with correspondence between RL components and neuroanatomy as follows: dopamine signal of substantia nigra pars compacta as the temporal difference error, striatum as the substrate for the critic, and the motor cortex as the actor. A key feature of this neurobiological interpretation is our hypothesis that the indirect pathway is the explorer. Chaotic activity, originating from the indirect pathway part of the model, drives the wandering, exploratory movements of the arm. Thus, the direct pathway subserves exploitation, while the indirect pathway subserves exploration. The motor cortex becomes more and more independent of the corrective influence of BG as training progresses. Reaching trajectories show diminishing variability with training. Reaching movements associated with Parkinson's disease (PD) are simulated by reducing dopamine and degrading the complexity of indirect pathway dynamics by switching it from chaotic to periodic behavior. Under the simulated PD conditions, the arm exhibits PD motor symptoms like tremor, bradykinesia and undershooting. The model echoes the notion that PD is a dynamical disease.

18. Conjugate Mixture Models for Clustering Multimodal Data

Vasil Khalidov, Florence Forbes, Radu Horaud

https://doi.org/10.1162/neco_a_00074 | pp. 517-557

The problem of multimodal clustering arises whenever the data are gathered with several physically different sensors. Observations from different modalities are not necessarily aligned in the sense there there is no obvious way to associate or compare them in some common space. A solution may consist in considering multiple clustering tasks independently for each modality. The main difficulty with such an approach is to guarantee that the unimodal clusterings are mutually consistent. In this letter, we show that multimodal clustering can be addressed within a novel framework: conjugate mixture models. These models exploit the explicit transformations that are often available between an unobserved parameter space (objects) and each of the observation spaces (sensors). We formulate the problem as a likelihood maximization task and derive the associated conjugate expectation-maximization algorithm. The convergence properties of the proposed algorithm are thoroughly investigated. Several local and global optimization techniques are proposed in order to increase its convergence speed. Two initialization strategies are proposed and compared. A consistent model selection criterion is proposed. The algorithm and its variants are tested and evaluated within the task of 3D localization of several speakers using both auditory and visual data.

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19. Learning Active Fusion of Multiple Experts' Decisions: An Attention-Based Approach

Maryam S. Mirian, Majid Nili Ahmadabadi, Babak Nadjar Araabi, Roland Siegwart

https://doi.org/10.1162/neco_a_00079 | pp. 558-591

In this letter, we propose a learning system, active decision fusion learning (ADFL), for active fusion of decisions. Each decision maker, referred to as a local decision maker, provides its suggestion in the form of a probability distribution over all possible decisions. The goal of the system is to learn the active sequential selection of the local decision makers in order to consult with and thus learn the final decision based on the consultations. These two learning tasks are formulated as learning a single sequential decision-making problem in the form of a Markov decision process (MDP), and a continuous reinforcement learning method is employed to solve it. The states of this MDP are decisions of the attended local decision makers, and the actions are either attending to a local decision maker or declaring final decisions. The learning system is punished for each consultation and wrong final decision and rewarded for correct final decisions. This results in minimizing the consultation and decision-making costs through learning a sequential consultation policy where the most informative local decision makers are consulted and the least informative, misleading, and redundant ones are left unattended. An important property of this policy is that it acts locally. This means that the system handles any nonuniformity in the local decision maker's expertise over the state space. This property has been exploited in the design of local experts. ADFL is tested on a set of classification tasks, where it outperforms two well-known classification methods, Adaboost and bagging, as well as three benchmark fusion algorithms: OWA, Borda count, and majority voting. In addition, the effect of local experts design strategy on the performance of ADFL is studied, and some guidelines for the design of local experts are provided. Moreover, evaluating ADFL in some special cases proves that it is able to derive the maximum benefit from the informative local decision makers and to minimize attending to redundant ones.

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20. Learning a Generative Model of Images by Factoring Appearance and Shape

Nicolas Le Roux, Nicolas Heess, Jamie Shotton, John Winn

https://doi.org/10.1162/neco_a_00086 | pp. 593-650

Computer vision has grown tremendously in the past two decades. Despite all efforts, existing attempts at matching parts of the human visual system's extraordinary ability to understand visual scenes lack either scope or power. By combining the advantages of general low-level generative models and powerful layer-based and hierarchical models, this work aims at being a first step toward richer, more flexible models of images. After comparing various types of restricted Boltzmann machines (RBMs) able to model continuous-valued data, we introduce our basic model, the masked RBM, which explicitly models occlusion boundaries in image patches by factoring the appearance of any patch region from its shape. We then propose a generative model of larger images using a field of such RBMs. Finally, we discuss how masked RBMs could be stacked to form a deep model able to generate more complicated structures and suitable for various tasks such as segmentation or object recognition.

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21. Intracellular Dynamics of Virtual Place Cells

Sandro Romani, Terrence J. Sejnowski, Misha Tsodyks

https://doi.org/10.1162/neco_a_00087 | pp. 651-655

The pattern of spikes recorded from place cells in the rodent hippocampus is strongly modulated by both the spatial location in the environment and the theta rhythm. The phases of the spikes in the theta cycle advance during movement through the place field. Recently intracellular recordings from hippocampal neurons (Harvey, Collman, Dombeck, & Tank, 2009) showed an increase in the amplitude of membrane potential oscillations inside the place field, which was interpreted as evidence that an intracellular mechanism caused phase precession. Here we show that an existing network model of the hippocampus (Tsodyks, Skaggs, Sejnowski, & McNaughton, 1996) can equally reproduce this and other aspects of the intracellular recordings, which suggests that new experiments are needed to distinguish the contributions of intracellular and network mechanisms to phase precession.

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22. Does High Firing Irregularity Enhance Learning?

Chris Christodoulou, Aristodemos Cleanthous

https://doi.org/10.1162/neco_a_00090 | pp. 656-663

In this note, we demonstrate that the high firing irregularity produced by the leaky integrate-and-fire neuron with the partial somatic reset mechanism, which has been shown to be the most likely candidate to reflect the mechanism used in the brain for reproducing the highly irregular cortical neuron firing at high rates (Bugmann, Christodoulou, & Taylor, 1997; Christodoulou & Bugmann, 2001), enhances learning. More specifically, it enhances reward-modulated spike-timing-dependent plasticity with eligibility trace when used in spiking neural networks, as shown by the results when tested in the simple benchmark problem of XOR, as well as in a complex multiagent setting task.

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23. Bounding the Bias of Contrastive Divergence Learning

Asja Fischer, Christian Igel

https://doi.org/10.1162/neco_a_00085 | pp. 664-673

Optimization based on k-step contrastive divergence (CD) has become a common way to train restricted Boltzmann machines (RBMs). The k-step CD is a biased estimator of the log-likelihood gradient relying on Gibbs sampling. We derive a new upper bound for this bias. Its magnitude depends on k, the number of variables in the RBM, and the maximum change in energy that can be produced by changing a single variable. The last reflects the dependence on the absolute values of the RBM parameters. The magnitude of the bias is also affected by the distance in variation between the modeled distribution and the starting distribution of the Gibbs chain.

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24. Stability Against Fluctuations: Scaling, Bifurcations, and Spontaneous Symmetry Breaking in Stochastic Models of Synaptic Plasticity

Terry Elliott

https://doi.org/10.1162/neco_a_00088 | pp. 674-734

In stochastic models of synaptic plasticity based on a random walk, the control of fluctuations is imperative. We have argued that synapses could act as low-pass filters, filtering plasticity induction steps before expressing a step change in synaptic strength. Earlier work showed, in simulation, that such a synaptic filter tames fluctuations very well, leading to patterns of synaptic connectivity that are stable for long periods of time. Here, we approach this problem analytically. We explicitly calculate the lifetime of meta-stable states of synaptic connectivity using a Fokker-Planck formalism in order to understand the dependence of this lifetime on both the plasticity step size and the filtering mechanism. We find that our analytical results agree very well with simulation results, despite having to make two approximations. Our analysis reveals, however, a deeper significance to the filtering mechanism and the plasticity step size. We show that a filter scales the step size into a smaller, effective step size. This scaling suggests that the step size may itself play the role of a temperature parameter, so that a filter cools the dynamics, thereby reducing the influence of fluctuations. Using the master equation, we explicitly demonstrate a bifurcation at a critical step size, confirming this interpretation. At this critical point, spontaneous symmetry breaking occurs in the class of stochastic models of synaptic plasticity that we consider.

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25. Collective Stability of Networks of Winner-Take-All Circuits

Ueli Rutishauser, Rodney J. Douglas, Jean-Jacques Slotine

https://doi.org/10.1162/neco_a_00091 | pp. 735-773

The neocortex has a remarkably uniform neuronal organization, suggesting that common principles of processing are employed throughout its extent. In particular, the patterns of connectivity observed in the superficial layers of the visual cortex are consistent with the recurrent excitation and inhibitory feedback required for cooperative-competitive circuits such as the soft winner-take-all (WTA). WTA circuits offer interesting computational properties such as selective amplification, signal restoration, and decision making. But these properties depend on the signal gain derived from positive feedback, and so there is a critical trade-off between providing feedback strong enough to support the sophisticated computations while maintaining overall circuit stability. The issue of stability is all the more intriguing when one considers that the WTAs are expected to be densely distributed through the superficial layers and that they are at least partially interconnected. We consider how to reason about stability in very large distributed networks of such circuits. We approach this problem by approximating the regular cortical architecture as many

interconnected cooperative-competitive modules. We demonstrate that by properly understanding the behavior of this small computational module, one can reason over the stability and convergence of very large networks composed of these modules. We obtain parameter ranges in which the WTA circuit operates in a high-gain regime, is stable, and can be aggregated arbitrarily to form large, stable networks. We use nonlinear contraction theory to establish conditions for stability in the fully nonlinear case and verify these solutions using numerical simulations. The derived bounds allow modes of operation in which the WTA network is multistable and exhibits state-dependent persistent activities. Our approach is sufficiently general to reason systematically about the stability of any network, biological or technological, composed of networks of small modules that express competition through shared inhibition.

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26. Suitability of V1 Energy Models for Object Classification

James Bergstra, Yoshua Bengio, Jérôme Louradour

https://doi.org/10.1162/neco_a_00084 | pp. 774-790

Simulations of cortical computation have often focused on networks built from simplified neuron models similar to rate models hypothesized for V1 simple cells. However, physiological research has revealed that even V1 simple cells have surprising complexity. Our computational simulations explore the effect of this complexity on the visual system's ability to solve simple tasks, such as the categorization of shapes and digits, after learning from a limited number of examples. We use recently proposed high-throughput methodology to explore what axes of modeling complexity are useful in these categorization tasks. We find that complex cell rate models learn to categorize objects better than simple cell models, and without incurring extra computational expense. We find that the squaring of linear filter responses leads to better performance. We find that several other components of physiologically derived models do not yield better performance.

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27. Machine-Learning-Based Coadaptive Calibration for Brain-Computer Interfaces

Carmen Vidaurre, Claudia Sannelli, Klaus■Robert Müller, Benjamin Blankertz

https://doi.org/10.1162/neco_a_00089 | pp. 791-816

Brain-computer interfaces (BCIs) allow users to control a computer application by brain activity as acquired (e.g., by EEG). In our classic machine learning approach to BCIs, the participants undertake a calibration measurement without feedback to acquire data to train the BCI system. After the training, the user can control a BCI and improve the operation through some type of feedback. However, not all BCI users are able to perform sufficiently well during feedback operation. In fact, a nonnegligible portion of participants (estimated 15%-30%) cannot control the system (a BCI illiteracy problem, generic to all motor-imagery-based BCIs). We hypothesize that one main difficulty for a BCI user is the transition from offline calibration to online feedback. In this work, we investigate adaptive machine learning methods to eliminate offline calibration and analyze the performance of 11 volunteers in a BCI based on the modulation of sensorimotor rhythms. We present an adaptation scheme that individually guides the user. It starts with a subject-independent classifier that evolves to a subject-optimized state-of-the-art classifier within one session while the user interacts continuously. These initial runs use supervised techniques for robust coadaptive learning of user and machine. Subsequent runs use unsupervised adaptation to track the

features' drift during the session and provide an unbiased measure of BCI performance. Using this approach, without any offline calibration, six users, including one novice, obtained good performance after 3 to 6 minutes of adaptation. More important, this novel guided learning also allows participants with BCI illiteracy to gain significant control with the BCI in less than 60 minutes. In addition, one volunteer without sensorimotor idle rhythm peak at the beginning of the BCI experiment developed it during the course of the session and used voluntary modulation of its amplitude to control the feedback application.

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28. Posterior Weighted Reinforcement Learning with State Uncertainty

Tobias Larsen, David S. Leslie, Edmund J. Collins, Rafa■ Bogacz

https://doi.org/10.1162/neco.2010.01-09-948 | pp. 1149-1179

Reinforcement learning models generally assume that a stimulus is presented that allows a learner to unambiguously identify the state of nature, and the reward received is drawn from a distribution that depends on that state. However, in any natural environment, the stimulus is noisy. When there is state uncertainty, it is no longer immediately obvious how to perform reinforcement learning, since the observed reward cannot be unambiguously allocated to a state of the environment. This letter addresses the problem of incorporating state uncertainty in reinforcement learning models. We show that simply ignoring the uncertainty and allocating the reward to the most likely state of the environment results in incorrect value estimates. Furthermore, using only the information that is available before observing the reward also results in incorrect estimates. We therefore introduce a new technique, posterior weighted reinforcement learning, in which the estimates of state probabilities are updated according to the observed rewards (e.g., if a learner observes a reward usually associated with a particular state, this state becomes more likely). We show analytically that this modified algorithm can converge to correct reward estimates and confirm this with numerical experiments. The algorithm is shown to be a variant of the expectation-maximization algorithm, allowing rigorous convergence analyses to be carried out. A possible neural implementation of the algorithm in the cortico-basal-ganglia-thalamic network is presented, and experimental predictions of our model are discussed.

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29. Nonconvergence in Logistic and Poisson Models for Neural Spiking

Mengyuan Zhao, Satish Iyengar

https://doi.org/10.1162/neco.2010.03-09-982 | pp. 1231-1244

Generalized linear models are an increasingly common approach for spike train data analysis. For the logistic and Poisson models, one possible difficulty is that iterative algorithms for computing parameter estimates may not converge because of certain data configurations. For the logistic model, these configurations are called complete and quasi-complete separation. We show that these features are likely to occur because of refractory periods of neurons. We use an example to study how standard software deals with this difficulty. For the Poisson model, we show that the same difficulties arise, this time possibly due to bursting or specifics of the binning. We characterize the nonconvergent configurations for both models, show that they can be detected by linear programming methods, and discuss possible remedies.

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30. A Gaussian Attractor Network for Memory and Recognition with Experience-Dependent Learning

Xiaolin Hu, Bo Zhang

https://doi.org/10.1162/neco.2010.02-09-957 | pp. 1333-1357

Attractor networks are widely believed to underlie the memory systems of animals across different species. Existing models have succeeded in qualitatively modeling properties of attractor dynamics, but their computational abilities often suffer from poor representations for realistic complex patterns, spurious attractors, low storage capacity, and difficulty in identifying attractive fields of attractors. We propose a simple two-layer architecture, gaussian attractor network, which has no spurious attractors if patterns to be stored are uncorrelated and can store as many patterns as the number of neurons in the output layer. Meanwhile the attractive fields can be precisely quantified and manipulated. Equipped with experience-dependent unsupervised learning strategies, the network can exhibit both discrete and continuous attractor dynamics. A testable prediction based on numerical simulations is that there exist neurons in the brain that can discriminate two similar stimuli at first but cannot after extensive exposure to physically intermediate stimuli. Inspired by this network, we found that adding some local feedbacks to a well-known hierarchical visual recognition model, HMAX, can enable the model to reproduce some recent experimental results related to high-level visual perception.

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31. Alteration of Ambient GABA by Phasic and Tonic Neuronal Activation

Osamu Hoshino

https://doi.org/10.1162/neco.2010.02-09-969 | pp. 1358-1382

Neurons of primary auditory cortex (AI) emit spikes (action potentials) in two distinct manners, responding to sounds in an onset or a sustained manner. The former AI neurons are called phasic cells and the latter tonic cells. The phasic cells generate spikes for a brief time period (less than hundreds of milliseconds) at the onset of an auditory stimulus (e.g., a tone frequency sound), and the tonic cells continuously generate spikes throughout the stimulation period. Simulating a neural network model of AI, we investigated whether and how the onset discharges influence the sustained discharges that are believed to play a central role in encoding auditory information. Onset discharges, triggered by a phasic input, briefly excited GABAergic interneurons and transiently increased the level of ambient GABA, which was immediately recognized by extrasynaptic GABA(a) receptors and provided inhibitory currents into neurons. The transient alteration of ambient GABA allowed tonic cells to respond selectively to a tonic input. The timing of phasic input relative to a tonic one had a great impact on the responsiveness of tonic cells. We found optimal timing for the best selective responsiveness: phasic input preceding tonic input by several tens of milliseconds. Offset discharges induced by a secondary input to phasic cells, applied at the end of the tonic input period, suddenly terminated the sustained discharges and allowed the network to return rapidly to the ongoing-spontaneous neuronal state. We suggest that the transporter-mediated alteration of ambient GABA, triggered by onset discharges, may improve the response property of AI neurons. Offset discharges may have a role in resetting AI neurons so that they can prepare for the next auditory input.

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32. Roles of Inhibitory Neurons in Rewiring-Induced Synchronization in Pulse-Coupled Neural Networks

Takashi Kanamaru, Kazuyuki Aihara
https://doi.org/10.1162/neco.2010.04-09-997 | pp. 1383-1398

The roles of inhibitory neurons in synchronous firing are examined in a network of excitatory and inhibitory neurons with Watts and Strogatz's rewiring. By examining the persistence of the synchronous firing that exists in the random network, it was found that there is a probability of rewiring at which a transition between the synchronous state and the asynchronous state takes place, and the dynamics of the inhibitory neurons play an important role in determining this probability.

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33. Reward-Modulated Hebbian Learning of Decision Making

Michael Pfeiffer, Bernhard Nessler, Rodney J. Douglas, Wolfgang Maass
https://doi.org/10.1162/neco.2010.03-09-980 | pp. 1399-1444

We introduce a framework for decision making in which the learning of decision making is reduced to its simplest and biologically most plausible form: Hebbian learning on a linear neuron. We cast our Bayesian-Hebb learning rule as reinforcement learning in which certain decisions are rewarded and prove that each synaptic weight will on average converge exponentially fast to the log-odd of receiving a reward when its pre- and postsynaptic neurons are active. In our simple architecture, a particular action is selected from the set of candidate actions by a winner-take-all operation. The global reward assigned to this action then modulates the update of each synapse. Apart from this global reward signal, our reward-modulated Bayesian Hebb rule is a pure Hebb update that depends only on the coactivation of the pre- and postsynaptic neurons, not on the weighted sum of all presynaptic inputs to the postsynaptic neuron as in the perceptron learning rule or the Rescorla-Wagner rule. This simple approach to action-selection learning requires that information about sensory inputs be presented to the Bayesian decision stage in a suitably preprocessed form resulting from other adaptive processes (acting on a larger timescale) that detect salient dependencies among input features. Hence our proposed framework for fast learning of decisions also provides interesting new hypotheses regarding neural nodes and computational goals of cortical areas that provide input to the final decision stage.

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34. General Poisson Exact Breakdown of the Mutual Information to Study the Role of Correlations in Populations of Neurons

Alessandro Scaglione, K. A. Moxon, Guglielmo Foffani
https://doi.org/10.1162/neco.2010.04-09-989 | pp. 1445-1467

We present an integrative formalism of mutual information expansion, the general Poisson exact breakdown, which explicitly evaluates the informational contribution of correlations in the spike counts both between and within neurons. The formalism was validated on simulated data and applied to real neurons recorded from the rat somatosensory cortex. From the general Poisson exact breakdown, a considerable number of mutual information measures introduced in the neural computation literature can be directly derived, including the exact breakdown (Pola, Thiele, Hoffmann, & Panzeri, 2003), the Poisson exact breakdown (Scaglione, Foffani, Scannella, Cerutti, & Moxon, 2008) the synergy and redundancy between neurons (Schneidman, Bialek, & Berry, 2003), and the information lost by an optimal decoder that assumes the absence of correlations between neurons (Nirenberg & Latham, 2003; Pola et al., 2003). The general

Poisson exact breakdown thus offers a convenient set of building blocks for studying the role of correlations in population codes.

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35. Fast and Exact Simulation Methods Applied on a Broad Range of Neuron Models

Michiel D'Haene, Benjamin Schrauwen

https://doi.org/10.1162/neco.2010.07-09-1070 | pp. 1468-1472

Recently van Elburg and van Ooyen (2009) published a generalization of the event-based integration scheme for an integrate-and-fire neuron model with exponentially decaying excitatory currents and double exponential inhibitory synaptic currents, introduced by Carnevale and Hines. In the paper, it was shown that the constraints on the synaptic time constants imposed by the Newton-Raphson iteration scheme, can be relaxed. In this note, we show that according to the results published in D'Haene, Schrauwen, Van Campenhout, and Stroobandt (2009), a further generalization is possible, eliminating any constraint on the time constants. We also demonstrate that in fact, a wide range of linear neuron models can be efficiently simulated with this computation scheme, including neuron models mimicking complex neuronal behavior. These results can change the way complex neuronal spiking behavior is modeled: instead of highly nonlinear neuron models with few state variables, it is possible to efficiently simulate linear models with a large number of state variables.

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36. Learning to Represent Spatial Transformations with Factored Higher-Order Boltzmann Machines

Roland Memisevic, Geoffrey E. Hinton

https://doi.org/10.1162/neco.2010.01-09-953 | pp. 1473-1492

To allow the hidden units of a restricted Boltzmann machine to model the transformation between two successive images, Memisevic and Hinton (2007) introduced three-way multiplicative interactions that use the intensity of a pixel in the first image as a multiplicative gain on a learned, symmetric weight between a pixel in the second image and a hidden unit. This creates cubically many parameters, which form a three-dimensional interaction tensor. We describe a low-rank approximation to this interaction tensor that uses a sum of factors, each of which is a three-way outer product. This approximation allows efficient learning of transformations between larger image patches. Since each factor can be viewed as an image filter, the model as a whole learns optimal filter pairs for efficiently representing transformations. We demonstrate the learning of optimal filter pairs from various synthetic and real image sequences. We also show how learning about image transformations allows the model to perform a simple visual analogy task, and we show how a completely unsupervised network trained on transformations perceives multiple motions of transparent dot patterns in the same way as humans.

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37. Timescale-Invariant Pattern Recognition by Feedforward Inhibition and Parallel Signal Processing

Felix Creutzig, Jan Benda, Sandra Wohlgemuth, Andreas Stumpner, Bernhard Ronacher, Andreas V. M. Herz

https://doi.org/10.1162/neco.2010.05-09-1016 | pp. 1493-1510

The timescale-invariant recognition of temporal stimulus sequences is vital for many species and poses a challenge for their sensory systems. Here we present a simple mechanistic model to address this computational task, based on recent observations in insects that use rhythmic acoustic communication signals for mate finding. In the model framework, feedforward inhibition leads to burst-like response patterns in one neuron of the circuit. Integrating these responses over a fixed time window by a readout neuron creates a timescale-invariant stimulus representation. Only two additional processing channels, each with a feature detector and a readout neuron, plus one final coincidence detector for all three parallel signal streams, are needed to account for the behavioral data. In contrast to previous solutions to the general time-warp problem, no time delay lines or sophisticated neural architectures are required. Our results suggest a new computational role for feedforward inhibition and underscore the power of parallel signal processing.

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38. Hyperbolically Discounted Temporal Difference Learning

William H. Alexander, Joshua W. Brown

https://doi.org/10.1162/neco.2010.08-09-1080 | pp. 1511-1527

Hyperbolic discounting of future outcomes is widely observed to underlie choice behavior in animals. Additionally, recent studies (Kobayashi & Schultz, 2008) have reported that hyperbolic discounting is observed even in neural systems underlying choice. However, the most prevalent models of temporal discounting, such as temporal difference learning, assume that future outcomes are discounted exponentially. Exponential discounting has been preferred largely because it can be expressed recursively, whereas hyperbolic discounting has heretofore been thought not to have a recursive definition. In this letter, we define a learning algorithm, hyperbolically discounted temporal difference (HDTD) learning, which constitutes a recursive formulation of the hyperbolic model.

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39. Response of Integrate-and-Fire Neurons to Noisy Inputs Filtered by Synapses with Arbitrary Timescales: Firing Rate and Correlations

Rubén Moreno■Bote, Néstor Parga

https://doi.org/10.1162/neco.2010.06-09-1036 | pp. 1528-1572

Delivery of neurotransmitter produces on a synapse a current that flows through the membrane and gets transmitted into the soma of the neuron, where it is integrated. The decay time of the current depends on the synaptic receptor's type and ranges from a few (e.g., AMPA receptors) to a few hundred milliseconds (e.g., NMDA receptors). The role of the variety of synaptic timescales, several of them coexisting in the same neuron, is at present not understood. A prime question to answer is which is the effect of temporal filtering at different timescales of the incoming spike trains on the neuron's response. Here, based on our previous work on linear synaptic filtering, we build a general theory for the stationary firing response of integrate-and-fire (IF) neurons receiving stochastic inputs filtered by one, two, or multiple synaptic channels, each characterized by an arbitrary timescale. The formalism applies to arbitrary IF model neurons and arbitrary forms of input noise (i.e., not required to be gaussian or to have small amplitude), as well as to any

form of synaptic filtering (linear or nonlinear). The theory determines with exact analytical expressions the firing rate of an IF neuron for long synaptic time constants using the adiabatic approach. The correlated spiking (cross-correlations function) of two neurons receiving common as well as independent sources of noise is also described. The theory is illustrated using leaky, quadratic, and noise-thresholded IF neurons. Although the adiabatic approach is exact when at least one of the synaptic timescales is long, it provides a good prediction of the firing rate even when the timescales of the synapses are comparable to that of the leak of the neuron; it is not required that the synaptic time constants are longer than the mean interspike intervals or that the noise has small variance. The distribution of the potential for general IF neurons is also characterized. Our results provide powerful analytical tools that can allow a quantitative description of the dynamics of neuronal networks with realistic synaptic dynamics.

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40. Bayesian Inference Based on Stationary Fokker-Planck Sampling

Arturo Berrones

https://doi.org/10.1162/neco.2010.01-09-943 | pp. 1573-1596

A novel formalism for bayesian learning in the context of complex inference models is proposed. The method is based on the use of the stationary Fokker-Planck (SFP) approach to sample from the posterior density. Stationary Fokker-Planck sampling generalizes the Gibbs sampler algorithm for arbitrary and unknown conditional densities. By the SFP procedure, approximate analytical expressions for the conditionals and marginals of the posterior can be constructed. At each stage of SFP, the approximate conditionals are used to define a Gibbs sampling process, which is convergent to the full joint posterior. By the analytical marginals efficient learning methods in the context of artificial neural networks are outlined. Offline and incremental bayesian inference and maximum likelihood estimation from the posterior are performed in classification and regression examples. A comparison of SFP with other Monte Carlo strategies in the general problem of sampling from arbitrary densities is also presented. It is shown that SFP is able to jump large low-probability regions without the need of a careful tuning of any step-size parameter. In fact, the SFP method requires only a small set of meaningful parameters that can be selected following clear, problem-independent guidelines. The computation cost of SFP, measured in terms of loss function evaluations, grows linearly with the given model's dimension.

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41. Efficient Continuous-Time Asymmetric Hopfield Networks for Memory Retrieval

Pengsheng Zheng, Wansheng Tang, Jianxiong Zhang

https://doi.org/10.1162/neco.2010.05-09-1014 | pp. 1597-1614

A novel m energy functions method is adopted to analyze the retrieval property of continuous-time asymmetric Hopfield neural networks. Sufficient conditions for the local and global asymptotic stability of the network are proposed. Moreover, an efficient systematic procedure for designing asymmetric networks is proposed, and a given set of states can be assigned as locally asymptotically stable equilibrium points. Simulation examples show that the asymmetric network can act as an efficient associative memory, and it is almost free from spurious memory problem.

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42. Learning the Dynamical System Behind Sensory Data

Jaehyung Lee, Soo Young Lee

https://doi.org/10.1162/neco.2010.01-09-941 | pp. 1615-1645

This letter looks at the physics behind sensory data by identifying the parameters that govern the physical system and estimating them from sensory observations. We extend Takens's delay-embedding theorem to the dynamical systems controlled by parameters. An embedding of the product space of the phase and the parameter spaces of the dynamical system can be obtained by the delay-embedding map, provided that the parameter of the dynamical system changes slowly. The reconstruction error is bounded for slowly varying parameters. A manifold learning technique is applied to the embedding obtained to extract a low-dimensional global coordinate system representing the product space. The phase space of the deterministic dynamics can be contracted by using the adjacency relationship in time, which enables recovery of only the parameter space. As examples, the manifolds of synthetic and real-world vowels with time-varying fundamental frequency (F(0)) are analyzed, and the F(0) contours are extracted by an unsupervised algorithm. Experimental results show that the proposed method leads to robust performance under various noise conditions and rapid changes of F(0) compared with the current state-of-the-art F(0) estimation algorithms.

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43. Independent Vector Analysis for Source Separation Using a Mixture of Gaussians Prior

Jiucang Hao, Intae Lee, Te-Won Lee, Terrence J. Sejnowski

https://doi.org/10.1162/neco.2010.11-08-906 | pp. 1646-1673

Convolutive mixtures of signals, which are common in acoustic environments, can be difficult to separate into their component sources. Here we present a uniform probabilistic framework to separate convolutive mixtures of acoustic signals using independent vector analysis (IVA), which is based on a joint distribution for the frequency components originating from the same source and is capable of preventing permutation disorder. Different gaussian mixture models (GMM) served as source priors, in contrast to the original IVA model, where all sources were modeled by identical multivariate Laplacian distributions. This flexible source prior enabled the IVA model to separate different type of signals. Three classes of models were derived and tested: noiseless IVA, online IVA, and noisy IVA. In the IVA model without sensor noise, the unmixing matrices were efficiently estimated by the expectation maximization (EM) algorithm. An online EM algorithm was derived for the online IVA algorithm to track the movement of the sources and separate them under nonstationary conditions. The noisy IVA model included the sensor noise and combined denoising with separation. An EM algorithm was developed that found the model parameters and separated the sources simultaneously. These algorithms were applied to separate mixtures of speech and music. Performance as measured by the signal-to-interference ratio (SIR) was substantial for all three models.

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44. First-Spike Latency in the Presence of Spontaneous Activity

Zbyn■k Pawlas, Lev B. Klebanov, Viktor Beneš, Michaela Prokešová, Ji■í Popelá■, Petr Lánský

https://doi.org/10.1162/neco.2010.11-09-1118 | pp. 1675-1697

A new statistical method for the estimation of the response latency is proposed. When spontaneous discharge is present, the first spike after the stimulus application may be caused by either the stimulus itself, or it may appear due to the prevailing spontaneous activity. Therefore, an appropriate method to deduce the response latency from the time to the first spike after the stimulus is needed. We develop a nonparametric estimator of the response latency based on repeated stimulations. A simulation study is provided to show how the estimator behaves with an increasing number of observations and for different rates of spontaneous and evoked spikes. Our nonparametric approach requires very few assumptions. For comparison, we also consider a parametric model. The proposed probabilistic model can be used for both single and parallel neuronal spike trains. In the case of simultaneously recorded spike trains in several neurons, the estimators of joint distribution and correlations of response latencies are also introduced. Real data from inferior colliculus auditory neurons obtained from a multielectrode probe are studied to demonstrate the statistical estimators of response latencies and their correlations in space.

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45. Learning Spike-Based Population Codes by Reward and Population Feedback

Johannes Friedrich, Robert Urbanczik, Walter Senn

https://doi.org/10.1162/neco.2010.05-09-1010 | pp. 1698-1717

We investigate a recently proposed model for decision learning in a population of spiking neurons where synaptic plasticity is modulated by a population signal in addition to reward feedback. For the basic model, binary population decision making based on spike/no-spike coding, a detailed computational analysis is given about how learning performance depends on population size and task complexity. Next, we extend the basic model to n-ary decision making and show that it can also be used in conjunction with other population codes such as rate or even latency coding.

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46. Conditional Mixture Model for Correlated Neuronal Spikes

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https://doi.org/10.1162/neco.2010.04-08-766 | pp. 1718-1736

Analysis of correlated spike trains is a hot topic of research in computational neuroscience. A general model of probability distributions for spikes includes too many parameters to be of use in analyzing real data. Instead, we need a simple but powerful generative model for correlated spikes. We developed a class of conditional mixture models that includes a number of existing models and analyzed its capabilities and limitations. We apply the model to dynamical aspects of neuron pools. When Hebbian cell assemblies coexist in a pool of neurons, the condition is specified by these assemblies such that the probability distribution of spikes is a mixture of those of the component assemblies. The probabilities of activation of the Hebbian assemblies change dynamically. We used this model as a basis for a competitive model governing the states of assemblies.

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47. On the Noise-Enhancing Ability of Stochastic Hodgkin-Huxley Neuron Systems

Bor■Sen Chen, Chengwei Li

https://doi.org/10.1162/neco.2010.07-09-1057 | pp. 1737-1763

Recently noise has been shown to be useful in enhancing neuron sensitivity by stochastic resonance. In this study, in order to measure the noise-enhancing factor (NEF), a nonlinear stochastic model is introduced for the Hodgkin-Huxley (HH) neuron system with synaptic noise input stimulation and channel noises in the sodium and potassium channels. The enhancing factor of the HH neuron system is measured from the point of view of the noise-exploiting level of nonlinear stochastic H(infinity) signal processing. Since the nonlinear stochastic-enhancing measure problem of HH neuron systems requires a solution for the difficulty presented by the Hamilton Jacobi inequality (HJI), a fuzzy interpolation of locally linearized systems is employed to simplify the nonlinear noise-enhancing problems by solving only a set of linear matrix inequalities. The NEF of the HH neuron system is found to be related to the locations of eigenvalues of linearized HH neuron systems and can be estimated through the H(infinity) signal processing method. Based on a stochastic fuzzy linearized HH neuron system, we found that channel noises are enhanced by the active eigenvalues of ionic channels while synaptic noises are attenuated by the passive eigenvalues of synaptic process.

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48. Statistical Computer Model Analysis of the Reciprocal and Recurrent Inhibitory Postsynaptic Potentials in α-Motoneurons

Gérard Gradwohl, Yoram Grossman

https://doi.org/10.1162/neco.2010.07-09-1054 | pp. 1764-1785

We simulate reconstructed alpha-motoneurons (MNs) under physiological and morphological realistic parameters and compare the modeled reciprocal (REC) and recurrent (REN) inhibitory postsynaptic potentials (IPSPs) containing voltage-dependent channels on the dendrites with the IPSPs of a passive MN model. Three distribution functions of the voltage-dependent channels on the dendrites are applied: a step function (ST) with uniform spatial dispersion; an exponential decay (ED) function, with channels with high density located proximal to the soma; and an exponential rise (ER) with a higher density of channels located distally. The excitatory and REN inhibitory inputs are located as a gaussian function on the dendrites, while the REC inhibitory synapses are located proximal to the soma. Our simulations generate four key results. (1) The distribution pattern of the voltage-dependent channels does not affect the IPSP peak, its time integral (TI), or its rate of rise (RR). However, the IPSP peak decreased in the presence of the active dendrites, while the EPSP peak increased. (2) Proximally located IPSP conductance produces greater IPSP peak, RR, and TI. (3) Increased duration of the IPSP produces greater RR and moderately increased TI and has a small effect on the peak amplitude. (4) The IPSP of both REC and REN models is specific to each MN: its amplitude is proportional to the MNs' input resistance, R(N); the increase of IPSP at the proximal location of the IPSP synapses is inversely related to R(N); and the effect of the IPSP conductance duration is insensitive to R(N).

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49. Decision Confidence and Uncertainty in Diffusion Models with Partially Correlated Neuronal Integrators

Rubén Moreno■Bote

https://doi.org/10.1162/neco.2010.12-08-930 | pp. 1786-1811

Diffusion models have become essential for describing the performance and statistics of reaction times in human decision making. Despite their success, it is not known how to evaluate decision confidence from them. I introduce a broader class of models consisting of two partially correlated neuronal integrators with arbitrarily time-varying decision boundaries that allow a natural description of confidence. The dependence of decision confidence on the state of the losing integrator, decision time, time-varying boundaries, and correlations is analytically described. The marginal confidence is computed for the half-anticorrelated case using the exact solution of the diffusion process with constant boundaries and compared to that of the independent and completely anticorrelated cases.

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50. Role of Homeostasis in Learning Sparse Representations

Laurent U. Perrinet

https://doi.org/10.1162/neco.2010.05-08-795 | pp. 1812-1836

Neurons in the input layer of primary visual cortex in primates develop edge-like receptive fields. One approach to understanding the emergence of this response is to state that neural activity has to efficiently represent sensory data with respect to the statistics of natural scenes. Furthermore, it is believed that such an efficient coding is achieved using a competition across neurons so as to generate a sparse representation, that is, where a relatively small number of neurons are simultaneously active. Indeed, different models of sparse coding, coupled with Hebbian learning and homeostasis, have been proposed that successfully match the observed emergent response. However, the specific role of homeostasis in learning such sparse representations is still largely unknown. By quantitatively assessing the efficiency of the neural representation during learning, we derive a cooperative homeostasis mechanism that optimally tunes the competition between neurons within the sparse coding algorithm. We apply this homeostasis while learning small patches taken from natural images and compare its efficiency with state-of-the-art algorithms. Results show that while different sparse coding algorithms give similar coding results, the homeostasis provides an optimal balance for the representation of natural images within the population of neurons. Competition in sparse coding is optimized when it is fair. By contributing to optimizing statistical competition across neurons, homeostasis is crucial in providing a more efficient solution to the emergence of independent components.

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51. Hebbian Plasticity and Homeostasis in a Model of Hypercolumn of the Visual Cortex

Román Rossi-Pool, Germán Mato

https://doi.org/10.1162/neco.2010.07-09-1056 | pp. 1837-1859

Neurons in the nervous system display a wide variety of plasticity processes. Among them are covariance-based rules and homeostatic plasticity. By themselves, the first ones tend to generate instabilities because of the unbounded potentiation of synapses. The second ones tend to stabilize the system by setting a target for the postsynaptic firing rate. In this work, we analyze the combined effect of these two mechanisms in a simple model of hypercolumn of the visual cortex. We find that the presence of homeostatic plasticity together with nonplastic uniform inhibition stabilizes the effect of Hebbian plasticity. The system can reach nontrivial solutions, where the recurrent intracortical connections are strongly modulated. The modulation is strong enough to generate contrast invariance. Moreover, this state can be

reached even beginning from a weakly modulated initial condition.

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52. Autonomous Evolution of Topographic Regularities in Artificial Neural Networks

Jason Gauci, Kenneth O. Stanley

https://doi.org/10.1162/neco.2010.06-09-1042 | pp. 1860-1898

Looking to nature as inspiration, for at least the past 25 years, researchers in the field of neuroevolution (NE) have developed evolutionary algorithms designed specifically to evolve artificial neural networks (ANNs). Yet the ANNs evolved through NE algorithms lack the distinctive characteristics of biological brains, perhaps explaining why NE is not yet a mainstream subject of neural computation. Motivated by this gap, this letter shows that when geometry is introduced to evolved ANNs through the hypercube-based neuroevolution of augmenting topologies algorithm, they begin to acquire characteristics that indeed are reminiscent of biological brains. That is, if the neurons in evolved ANNs are situated at locations in space (i.e., if they are given coordinates), then, as experiments in evolving checkers-playing ANNs in this letter show, topographic maps with symmetries and regularities can evolve spontaneously. The ability to evolve such maps is shown in this letter to provide an important advantage in generalization. In fact, the evolved maps are sufficiently informative that their analysis yields the novel insight that the geometry of the connectivity patterns of more general players is significantly smoother and more contiguous than less general ones. Thus, the results reveal a correlation between generality and smoothness in connectivity patterns. They also hint at the intriguing possibility that as NE matures as a field, its algorithms can evolve ANNs of increasing relevance to those who study neural computation in general.

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53. Memory Dynamics in Attractor Networks with Saliency Weights

Huajin Tang, Haizhou Li, Rui Yan

https://doi.org/10.1162/neco.2010.07-09-1050 | pp. 1899-1926

Memory is a fundamental part of computational systems like the human brain. Theoretical models identify memories as attractors of neural network activity patterns based on the theory that attractor (recurrent) neural networks are able to capture some crucial characteristics of memory, such as encoding, storage, retrieval, and long-term and working memory. In such networks, long-term storage of the memory patterns is enabled by synaptic strengths that are adjusted according to some activity-dependent plasticity mechanisms (of which the most widely recognized is the Hebbian rule) such that the attractors of the network dynamics represent the stored memories. Most of previous studies on associative memory are focused on Hopfield-like binary networks, and the learned patterns are often assumed to be uncorrelated in a way that minimal interactions between memories are facilitated. In this letter, we restrict our attention to a more biological plausible attractor network model and study the neuronal representations of correlated patterns. We have examined the role of saliency weights in memory dynamics. Our results demonstrate that the retrieval process of the memorized patterns is characterized by the saliency distribution, which affects the landscape of the attractors. We have established the conditions that the network state converges to unique memory and multiple memories. The analytical result also holds for other cases for variable coding levels and nonbinary levels, indicating a general property emerging from correlated memories. Our results confirmed the advantage of computing with graded-response neurons over binary neurons (i.e., reducing of

spurious states). It was also found that the nonuniform saliency distribution can contribute to disappearance of spurious states when they exit.

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54. Norm-Observable Operator Models

Mingjie Zhao, Herbert Jaeger

https://doi.org/10.1162/neco.2010.03-09-983 | pp. 1927-1959

Hidden Markov models (HMMs) are one of the most popular and successful statistical models for time series. Observable operator models (OOMs) are generalizations of HMMs that exhibit several attractive advantages. In particular, a variety of highly efficient, constructive, and asymptotically correct learning algorithms are available for OOMs. However, the OOM theory suffers from the negative probability problem (NPP): a given, learned OOM may sometimes predict negative probabilities for certain events. It was recently shown that it is undecidable whether a given OOM will eventually produce such negative values. We propose a novel variant of OOMs, called norm-observable operator models (NOOMs), which avoid the NPP by design. Like OOMs, NOOMs use a set of linear operators to update system states. But differing from OOMs, they represent probabilities by the square of the norm of system states, thus precluding negative probability values. While being free of the NPP, NOOMs retain most advantages of OOMs. For example, NOOMs also capture (some) processes that cannot be modeled by HMMs. More importantly, in principle, NOOMs can be learned from data in a constructive way, and the learned models are asymptotically correct. We also prove that NOOMs capture all Markov chain (MC) describable processes. This letter presents the mathematical foundations of NOOMs, discusses the expressiveness of the model class, and explains how a NOOM can be estimated from data constructively.

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55. A Spiking Neuron as Information Bottleneck

Lars Buesing, Wolfgang Maass

https://doi.org/10.1162/neco.2010.08-09-1084 | pp. 1961-1992

Neurons receive thousands of presynaptic input spike trains while emitting a single output spike train. This drastic dimensionality reduction suggests considering a neuron as a bottleneck for information transmission. Extending recent results, we propose a simple learning rule for the weights of spiking neurons derived from the information bottleneck (IB) framework that minimizes the loss of relevant information transmitted in the output spike train. In the IB framework, relevance of information is defined with respect to contextual information, the latter entering the proposed learning rule as a "third" factor besides pre- and postsynaptic activities. This renders the theoretically motivated learning rule a plausible model for experimentally observed synaptic plasticity phenomena involving three factors. Furthermore, we show that the proposed IB learning rule allows spiking neurons to learn a predictive code, that is, to extract those parts of their input that are predictive for future input.

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56. Estimating a State-Space Model from Point Process Observations: A Note on Convergence

Ke Yuan, Mahesan Niranjan

https://doi.org/10.1162/neco.2010.07-09-1047 | pp. 1993-2001

Physiological signals such as neural spikes and heartbeats are discrete events in time, driven by continuous underlying systems. A recently introduced data-driven model to analyze such a system is a state-space model with point process observations, parameters of which and the underlying state sequence are simultaneously identified in a maximum likelihood setting using the expectation-maximization (EM) algorithm. In this note, we observe some simple convergence properties of such a setting, previously un-noticed. Simulations show that the likelihood is unimodal in the unknown parameters, and hence the EM iterations are always able to find the globally optimal solution.

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57. A Computationally Efficient Method for Nonparametric Modeling of Neural Spiking Activity with Point Processes

Todd P. Coleman, Sridevi S. Sarma

https://doi.org/10.1162/neco_a_00001-coleman | pp. 2002-2030

Point-process models have been shown to be useful in characterizing neural spiking activity as a function of extrinsic and intrinsic factors. Most point-process models of neural activity are parametric, as they are often efficiently computable. However, if the actual point process does not lie in the assumed parametric class of functions, misleading inferences can arise. Nonparametric methods are attractive due to fewer assumptions, but computation in general grows with the size of the data. We propose a computationally efficient method for nonparametric maximum likelihood estimation when the conditional intensity function, which characterizes the point process in its entirety, is assumed to be a Lipschitz continuous function but otherwise arbitrary. We show that by exploiting much structure, the problem becomes efficiently solvable. We next demonstrate a model selection procedure to estimate the Lipshitz parameter from data, akin to the minimum description length principle and demonstrate consistency of our estimator under appropriate assumptions. Finally, we illustrate the effectiveness of our method with simulated neural spiking data, goldfish retinal ganglion neural data, and activity recorded in CA1 hippocampal neurons from an awake behaving rat. For the simulated data set, our method uncovers a more compact representation of the conditional intensity function when it exists. For the goldfish and rat neural data sets, we show that our nonparametric method gives a superior absolute goodness-of-fit measure used for point processes than the most common parametric and splines-based approaches.

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58. How Synaptic Release Probability Shapes Neuronal Transmission: Information-Theoretic Analysis in a Cerebellar Granule Cell

Angelo Arleo, Thierry Nieus, Michele Bezzi, Anna D'Errico, Egidio D'Angelo, Olivier J. M. D. Coenen

https://doi.org/10.1162/neco_a_00006-arleo | pp. 2031-2058

A nerve cell receives multiple inputs from upstream neurons by way of its synapses. Neuron processing functions are thus influenced by changes in the biophysical properties of the synapse, such as long-term potentiation (LTP) or depression (LTD). This observation has opened new perspectives on the biophysical basis of learning and memory, but its quantitative impact on the information transmission of a neuron remains partially elucidated. One major obstacle is the high dimensionality of the neuronal input-output space, which makes it unfeasible to perform a thorough computational analysis of a neuron with multiple

synaptic inputs. In this work, information theory was employed to characterize the information transmission of a cerebellar granule cell over a region of its excitatory input space following synaptic changes. Granule cells have a small dendritic tree (on average, they receive only four mossy fiber afferents), which greatly bounds the input combinatorial space, reducing the complexity of information-theoretic calculations. Numerical simulations and LTP experiments quantified how changes in neurotransmitter release probability (p) modulated information transmission of a cerebellar granule cell. Numerical simulations showed that p shaped the neurotransmission landscape in unexpected ways. As p increased, the optimality of the information transmission of most stimuli did not increase strictly monotonically; instead it reached a plateau at intermediate p levels. Furthermore, our results showed that the spatiotemporal characteristics of the inputs determine the effect of p on neurotransmission, thus permitting the selection of distinctive preferred stimuli for different p values. These selective mechanisms may have important consequences on the encoding of cerebellar mossy fiber inputs and the plasticity and computation at the next circuit stage, including the parallel fiber-Purkinje cell synapses.

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59. Reconciling the STDP and BCM Models of Synaptic Plasticity in a Spiking Recurrent Neural Network

Daniel Bush, Andrew Philippides, Phil Husbands, Michael O'Shea

https://doi.org/10.1162/neco_a_00003-bush | pp. 2059-2085

Rate-coded Hebbian learning, as characterized by the BCM formulation, is an established computational model of synaptic plasticity. Recently it has been demonstrated that changes in the strength of synapses in vivo can also depend explicitly on the relative timing of pre- and postsynaptic firing. Computational modeling of this spike-timing-dependent plasticity (STDP) has demonstrated that it can provide inherent stability or competition based on local synaptic variables. However, it has also been demonstrated that these properties rely on synaptic weights being either depressed or unchanged by an increase in mean stochastic firing rates, which directly contradicts empirical data. Several analytical studies have addressed this apparent dichotomy and identified conditions under which distinct and disparate STDP rules can be reconciled with rate-coded Hebbian learning. The aim of this research is to verify, unify, and expand on these previous findings by manipulating each element of a standard computational STDP model in turn. This allows us to identify the conditions under which this plasticity rule can replicate experimental data obtained using both rate and temporal stimulation protocols in a spiking recurrent neural network. Our results describe how the relative scale of mean synaptic weights and their dependence on stochastic pre- or postsynaptic firing rates can be manipulated by adjusting the exact profile of the asymmetric learning window and temporal restrictions on spike pair interactions respectively. These findings imply that previously disparate models of rate-coded autoassociative learning and temporally coded heteroassociative learning, mediated by symmetric and asymmetric connections respectively, can be implemented in a single network using a single plasticity rule. However, we also demonstrate that forms of STDP that can be reconciled with rate-coded Hebbian learning do not generate inherent synaptic competition, and thus some additional mechanism is required to guarantee long-term input-output selectivity.

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60. Multilayer Processing of Spatiotemporal Spike Patterns in a Neuron with Active Dendrites

Yingxue Wang, Shih-Chii Liu

https://doi.org/10.1162/neco.2010.06-09-1030 | pp. 2086-2112

With the advent of new experimental evidence showing that dendrites play an active role in processing a neuron's inputs, we revisit the question of a suitable abstraction for the computing function of a neuron in processing spatiotemporal input patterns. Although the integrative role of a neuron in relation to the spatial clustering of synaptic inputs can be described by a two-layer neural network, no corresponding abstraction has yet been described for how a neuron processes temporal input patterns on the dendrites. We address this void using a real-time aVLSI (analog very-large-scale-integrated) dendritic compartmental model, which incorporates two widely studied classes of regenerative event mechanisms: one is mediated by voltage-gated ion channels and the other by transmitter-gated NMDA channels. From this model, we find that the response of a dendritic compartment can be described as a nonlinear sigmoidal function of both the degree of input temporal synchrony and the synaptic input spatial clustering. We propose that a neuron with active dendrites can be modeled as a multilayer network that selectively amplifies responses to relevant spatiotemporal input spike patterns.

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61. Spectral Method and High-Order Finite Differences for the Nonlinear Cable Equation

Ahmet Omurtag, William W. Lytton

https://doi.org/10.1162/neco.2010.09-09-1097 | pp. 2113-2136

We use high-order approximation schemes for the space derivatives in the nonlinear cable equation and investigate the behavior of numerical solution errors by using exact solutions, where available, and grid convergence. The space derivatives are numerically approximated by means of differentiation matrices. Nonlinearity in the equation arises from the Hodgkin-Huxley dynamics of the gating variables for ion channels. We have investigated in particular the effects of synaptic current distribution and compared the accuracy of the spectral solutions with that of finite differencing. A flexible form for the injected current is used that can be adjusted smoothly from a very broad to a narrow peak, which furthermore leads, for the passive cable, to a simple, exact solution. We have used three distinct approaches to assess the numerical solutions: comparison with exact solutions in an unbranched passive cable, the convergence of solutions with progressive refinement of the grid in an active cable, and the simulation of spike initiation in a biophysically realistic single-neuron model. The spectral method provides good numerical solutions for passive cables comparable in accuracy to those from the second-order finite difference method and far greater accuracy in the case of a simulated system driven by inputs that are smoothly distributed in space. It provides faster convergence in active cables and in a realistic neuron model due to better approximation of propagating spikes.

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62. Competitive Layer Model of Discrete-Time Recurrent Neural Networks with LT Neurons

Wei Zhou, Jacek M. ■urada

https://doi.org/10.1162/neco_a_00004-zhou | pp. 2137-2160

This letter discusses the competitive layer model (CLM) for a class of discrete-time recurrent neural networks with linear threshold (LT) neurons. It first addresses the boundedness, global attractivity, and complete stability of the networks. Two theorems are then presented for the networks to have CLM property. We also present the analysis for network dynamics, which performs a column winner-take-all behavior and grouping selection among different layers. Furthermore, we propose a novel synchronous CLM iteration method, which has similar performance and storage allocation but faster convergence compared with the previous asynchronous CLM iteration method (Wersing, Steil, & Ritter, 2001). Examples and simulation results are used to illustrate the developed theory, the comparison between two CLM iteration methods, and the application in image segmentation.

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63. Modeling Stereopsis via Markov Random Field

Yansheng Ming, Zhanyi Hu

https://doi.org/10.1162/neco_a_00005-ming | pp. 2161-2191

Markov random field (MRF) and belief propagation have given birth to stereo vision algorithms with top performance. This article explores their biological plausibility. First, an MRF model guided by physiological and psychophysical facts was designed. Typically an MRF-based stereo vision algorithm employs a likelihood function that reflects the local similarity of two regions and a potential function that models the continuity constraint. In our model, the likelihood function is constructed on the basis of the disparity energy model because complex cells are considered as front-end disparity encoders in the visual pathway. Our likelihood function is also relevant to several psychological findings. The potential function in our model is constrained by the psychological finding that the strength of the cooperative interaction minimizing relative disparity decreases as the separation between stimuli increases. Our model is tested on three kinds of stereo images. In simulations on images with repetitive patterns, we demonstrate that our model could account for the human depth percepts that were previously explained by the second-order mechanism. In simulations on random dot stereograms and natural scene images, we demonstrate that false matches introduced by the disparity energy model can be reliably removed using our model. A comparison with the coarse-to-fine model shows that our model is able to compute the absolute disparity of small objects with larger relative disparity. We also relate our model to several physiological findings. The hypothesized neurons of the model are selective for absolute disparity and have facilitative extra receptive field. There are plenty of such neurons in the visual cortex. In conclusion, we think that stereopsis can be implemented by neural networks resembling MRF.

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64. Deep Belief Networks Are Compact Universal Approximators

Nicolas Le Roux, Yoshua Bengio

https://doi.org/10.1162/neco.2010.08-09-1081 | pp. 2192-2207

Deep belief networks (DBN) are generative models with many layers of hidden causal variables, recently introduced by Hinton, Osindero, and Teh (2006), along with a greedy layer-wise unsupervised learning algorithm. Building on Le Roux and Bengio (2008) and Sutskever and Hinton (2008), we show that deep but narrow generative networks do not require more parameters than shallow ones to achieve universal approximation. Exploiting the proof technique, we prove that deep but narrow feedforward neural networks with sigmoidal units can represent any Boolean expression.

65. Sample-Spacings-Based Density and Entropy Estimators for Spherically Invariant Multidimensional Data

Intae Lee

https://doi.org/10.1162/neco.2010.02-09-972 | pp. 2208-2227

While the sample-spacings-based density estimation method is simple and efficient, its applicability has been restricted to one-dimensional data. In this letter, the method is generalized such that it can be extended to multiple dimensions in certain circumstances. As a consequence, a multidimensional entropy estimator of spherically invariant continuous random variables is derived. Partial bias of the estimator is analyzed, and the estimator is further used to derive a nonparametric objective function for frequency-domain independent component analysis. The robustness and the effectiveness of the objective function are demonstrated with simulation results.

66. Topographic Mapping of Large Dissimilarity Data Sets

Barbara Hammer, Alexander Hasenfuß

https://doi.org/10.1162/neco_a_00012 | pp. 2229-2284

Topographic maps such as the self-organizing map (SOM) or neural gas (NG) constitute powerful data mining techniques that allow simultaneously clustering data and inferring their topological structure, such that additional features, for example, browsing, become available. Both methods have been introduced for vectorial data sets; they require a classical feature encoding of information. Often data are available in the form of pairwise distances only, such as arise from a kernel matrix, a graph, or some general dissimilarity measure. In such cases, NG and SOM cannot be applied directly. In this article, we introduce relational topographic maps as an extension of relational clustering algorithms, which offer prototype-based representations of dissimilarity data, to incorporate neighborhood structure. These methods are equivalent to the standard (vectorial) techniques if a Euclidean embedding exists, while preventing the need to explicitly compute such an embedding. Extending these techniques for the general case of non-Euclidean dissimilarities makes possible an interpretation of relational clustering as clustering in pseudo-Euclidean space. We compare the methods to well-known clustering methods for proximity data based on deterministic annealing and discuss how far convergence can be guaranteed in the general case. Relational clustering is quadratic in the number of data points, which makes the algorithms infeasible for huge data sets. We propose an approximate patch version of relational clustering that runs in linear time. The effectiveness of the methods is demonstrated in a number of examples.

67. Tractable Multivariate Binary Density Estimation and the Restricted Boltzmann Forest

Hugo Larochelle, Yoshua Bengio, Joseph Turian

https://doi.org/10.1162/neco_a_00014 | pp. 2285-2307

We investigate the problem of estimating the density function of multivariate binary data. In particular, we focus on models for which computing the estimated probability of any data point is tractable. In such a setting, previous work has mostly concentrated on mixture modeling approaches. We argue that for the

problem of tractable density estimation, the restricted Boltzmann machine (RBM) provides a competitive framework for multivariate binary density modeling. With this in mind, we also generalize the RBM framework and present the restricted Boltzmann forest (RBForest), which replaces the binary variables in the hidden layer of RBMs with groups of tree-structured binary variables. This extension allows us to obtain models that have more modeling capacity but remain tractable. In experiments on several data sets, we demonstrate the competitiveness of this approach and study some of its properties.

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68. A Two-Layer Model of Natural Stimuli Estimated with Score Matching

Urs Köster, Aapo Hyvärinen

https://doi.org/10.1162/neco_a_00010 | pp. 2308-2333

We consider a hierarchical two-layer model of natural signals in which both layers are learned from the data. Estimation is accomplished by score matching, a recently proposed estimation principle for energy-based models. If the first-layer outputs are squared and the second-layer weights are constrained to be nonnegative, the model learns responses similar to complex cells in primary visual cortex from natural images. The second layer pools a small number of features with similar orientation and frequency, but differing in spatial phase. For speech data, we obtain analogous results. The model unifies previous extensions to independent component analysis such as subspace and topographic models and provides new evidence that localized, oriented, phase-invariant features reflect the statistical properties of natural image patches.

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69. Individual Differences in Nucleus Accumbens Dopamine Receptors Predict Development of Addiction-Like Behavior: A Computational Approach

Payam Piray, Mohammad Mahdi Keramati, Amir Dezfouli, Caro Lucas, Azarakhsh Mokri

https://doi.org/10.1162/neco_a_00009 | pp. 2334-2368

Clinical and experimental observations show individual differences in the development of addiction. Increasing evidence supports the hypothesis that dopamine receptor availability in the nucleus accumbens (NAc) predisposes drug reinforcement. Here, modeling striatal-midbrain dopaminergic circuit, we propose a reinforcement learning model for addiction based on the actor-critic model of striatum. Modeling dopamine receptors in the NAc as modulators of learning rate for appetitive--but not aversive--stimuli in the critic--but not the actor--we define vulnerability to addiction as a relatively lower learning rate for the appetitive stimuli, compared to aversive stimuli, in the critic. We hypothesize that an imbalance in this learning parameter used by appetitive and aversive learning systems can result in addiction. We elucidate that the interaction between the degree of individual vulnerability and the duration of exposure to drug has two progressive consequences: deterioration of the imbalance and establishment of an abnormal habitual response in the actor. Using computational language, the proposed model describes how development of compulsive behavior can be a function of both degree of drug exposure and individual vulnerability. Moreover, the model describes how involvement of the dorsal striatum in addiction can be augmented progressively. The model also interprets other forms of addiction, such as obesity and pathological gambling, in a common mechanism with drug addiction. Finally, the model provides an answer for the question of why behavioral addictions are triggered in Parkinson's disease patients by D2 dopamine agonist treatments.

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70. Extracting State Transition Dynamics from Multiple Spike Trains Using Hidden Markov Models with Correlated Poisson Distribution

Kentaro Katahira, Jun Nishikawa, Kazuo Okanoya, Masato Okada

https://doi.org/10.1162/neco.2010.08-08-838 | pp. 2369-2389

Neural activity is nonstationary and varies across time. Hidden Markov models (HMMs) have been used to track the state transition among quasi-stationary discrete neural states. Within this context, an independent Poisson model has been used for the output distribution of HMMs; hence, the model is incapable of tracking the change in correlation without modulating the firing rate. To achieve this, we applied a multivariate Poisson distribution with correlation terms for the output distribution of HMMs. We formulated a variational Bayes (VB) inference for the model. The VB could automatically determine the appropriate number of hidden states and correlation types while avoiding the overlearning problem. We developed an efficient algorithm for computing posteriors using the recursive relationship of a multivariate Poisson distribution. We demonstrated the performance of our method on synthetic data and real spike trains recorded from a songbird.

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71. Sound Retrieval and Ranking Using Sparse Auditory Representations

Richard F. Lyon, Martin Rehn, Samy Bengio, Thomas C. Walters, Gal Chechik

https://doi.org/10.1162/neco_a_00011 | pp. 2390-2416

To create systems that understand the sounds that humans are exposed to in everyday life, we need to represent sounds with features that can discriminate among many different sound classes. Here, we use a sound-ranking framework to quantitatively evaluate such representations in a large-scale task. We have adapted a machine-vision method, the passive-aggressive model for image retrieval (PAMIR), which efficiently learns a linear mapping from a very large sparse feature space to a large query-term space. Using this approach, we compare different auditory front ends and different ways of extracting sparse features from high-dimensional auditory images. We tested auditory models that use an adaptive pole-zero filter cascade (PZFC) auditory filter bank and sparse-code feature extraction from stabilized auditory images with multiple vector quantizers. In addition to auditory image models, we compare a family of more conventional mel-frequency cepstral coefficient (MFCC) front ends. The experimental results show a significant advantage for the auditory models over vector-quantized MFCCs. When thousands of sound files with a query vocabulary of thousands of words were ranked, the best precision at top-1 was 73% and the average precision was 35%, reflecting a 18% improvement over the best competing MFCC front end.

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72. A Grouped Ranking Model for Item Preference Parameter

Hideitsu Hino, Yu Fujimoto, Noboru Murata

https://doi.org/10.1162/neco_a_00008 | pp. 2417-2451

Given a set of rating data for a set of items, determining preference levels of items is a matter of importance. Various probability models have been proposed to solve this task. One such model is the Plackett-Luce model, which parameterizes the preference level of each item by a real value. In this letter, the Plackett-Luce model is generalized to cope with grouped ranking observations such as movie or restaurant ratings. Since it is difficult to maximize the likelihood of the proposed model directly, a feasible

approximation is derived, and the em algorithm is adopted to find the model parameter by maximizing the approximate likelihood which is easily evaluated. The proposed model is extended to a mixture model, and two applications are proposed. To show the effectiveness of the proposed model, numerical experiments with real-world data are carried out.

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73. Bayesian Online Learning of the Hazard Rate in Change-Point Problems

Robert C. Wilson, Matthew R. Nassar, Joshua I. Gold

https://doi.org/10.1162/neco_a_00007 | pp. 2452-2476

Change-point models are generative models of time-varying data in which the underlying generative parameters undergo discontinuous changes at different points in time known as change points. Change-points often represent important events in the underlying processes, like a change in brain state reflected in EEG data or a change in the value of a company reflected in its stock price. However, change-points can be difficult to identify in noisy data streams. Previous attempts to identify change-points online using Bayesian inference relied on specifying in advance the rate at which they occur, called the hazard rate (h). This approach leads to predictions that can depend strongly on the choice of h and is unable to deal optimally with systems in which h is not constant in time. In this letter, we overcome these limitations by developing a hierarchical extension to earlier models. This approach allows h itself to be inferred from the data, which in turn helps to identify when change-points occur. We show that our approach can effectively identify change-points in both toy and real data sets with complex hazard rates and how it can be used as an ideal-observer model for human and animal behavior when faced with rapidly changing inputs.

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74. Discrete Time Rescaling Theorem: Determining Goodness of Fit for Discrete Time Statistical Models of Neural Spiking

Robert Haslinger, Gordon Pipa, Emery N. Brown

https://doi.org/10.1162/neco_a_00015 | pp. 2477-2506

One approach for understanding the encoding of information by spike trains is to fit statistical models and then test their goodness of fit. The time-rescaling theorem provides a goodness-of-fit test consistent with the point process nature of spike trains. The interspike intervals (ISIs) are rescaled (as a function of the model's spike probability) to be independent and exponentially distributed if the model is accurate. A Kolmogorov-Smirnov (KS) test between the rescaled ISIs and the exponential distribution is then used to check goodness of fit. This rescaling relies on assumptions of continuously defined time and instantaneous events. However, spikes have finite width, and statistical models of spike trains almost always discretize time into bins. Here we demonstrate that finite temporal resolution of discrete time models prevents their rescaled ISIs from being exponentially distributed. Poor goodness of fit may be erroneously indicated even if the model is exactly correct. We present two adaptations of the time-rescaling theorem to discrete time models. In the first we propose that instead of assuming the rescaled times to be exponential, the reference distribution be estimated through direct simulation by the fitted model. In the second, we prove a discrete time version of the time-rescaling theorem that analytically corrects for the effects of finite resolution. This allows us to define a rescaled time that is exponentially distributed, even at arbitrary temporal discretizations. We demonstrate the efficacy of both techniques by fitting generalized linear models to both simulated spike trains and spike trains recorded experimentally in monkey V1 cortex. Both techniques give

nearly identical results, reducing the false-positive rate of the KS test and greatly increasing the reliability of model evaluation based on the time-rescaling theorem.

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75. Convergence and Stability of Quantized Hopfield Networks Operating in a Fully Parallel Mode

Daniel Calabuig, José F. Monserrat, Narcís Cardona

https://doi.org/10.1162/neco_a_00002-calabuig | pp. 2507-2521

Hopfield neural networks (HNNs) have proven useful in solving optimization problems that require fast response times. However, the original analog model has an extremely high implementation complexity, making discrete implementations more suitable. Previous work has studied the convergence of discrete-time and quantized-neuron models but has limited the analysis to either two-state neurons or serial operation mode. Nevertheless, two-state neurons have poor performance, and serial operation modes lose fast convergence, which is characteristic of analog HNNs. This letter is the first in the field analyzing the convergence and stability of quantized Hopfield networks (QHNs)-with more than two states-operating in fully parallel mode. Moreover, this letter presents some further analysis on the energy minimization of this type of network. The main conclusion drawn is that QHNs operating in fully parallel mode always converge to a stable state or a cycle of length two and any stable state is a local minimum of the energy.

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76. Probability of Repeating Patterns in Simultaneous Neural Data

Anne C. Smith, Vinh Nguyen, Mattias Karlsson, Loren M. Frank, Peter Smith

https://doi.org/10.1162/neco_a_00020 | pp. 2522-2536

Large data sets arising from neurophysiological experiments are frequently observed with repeating temporal patterns. Our ability to decode these patterns is dependent on the development of methods to assess whether the patterns are significant or occurring by chance. Given a hypothesized sequence within these data, we derive probability formulas to allow assessment of the likelihood of recurrence occurring by chance. We illustrate our approach using data from hippocampal neurons from awake, behaving rats.

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77. Recording from Two Neurons: Second-Order Stimulus Reconstruction from Spike Trains and Population Coding

Nelson Fernandes, B. D. L. Pinto, Lírio Onofre Baptista de Almeida, J. F. W. Slaets, R. Köberle

https://doi.org/10.1162/neco_a_00016 | pp. 2537-2557

We study the reconstruction of visual stimuli from spike trains, representing the reconstructed stimulus by a Volterra series up to second order. We illustrate this procedure in a prominent example of spiking neurons, recording simultaneously from the two H1 neurons located in the lobula plate of the fly Chrysomya megacephala. The fly views two types of stimuli, corresponding to rotational and translational displacements. Second-order reconstructions require the manipulation of potentially very large matrices, which obstructs the use of this approach when there are many neurons. We avoid the computation and inversion of these matrices using a convenient set of basis functions to expand our variables in. This requires approximating the spike train four-point functions by combinations of two-point functions similar to

relations, which would be true for gaussian stochastic processes. In our test case, this approximation does not reduce the quality of the reconstruction. The overall contribution to stimulus reconstruction of the second-order kernels, measured by the mean squared error, is only about 5% of the first-order contribution. Yet at specific stimulus-dependent instants, the addition of second-order kernels represents up to 100% improvement, but only for rotational stimuli. We present a perturbative scheme to facilitate the application of our method to weakly correlated neurons.

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78. On a Stochastic Leaky Integrate-and-Fire Neuronal Model

Aniello Buonocore, L. Caputo, Enrica Pirozzi, Luigi M. Ricciardi

https://doi.org/10.1162/neco_a_00023 | pp. 2558-2585

The leaky integrate-and-fire neuronal model proposed in Stevens and Zador (1998), in which time constant and resting potential are postulated to be time dependent, is revisited within a stochastic framework in which the membrane potential is mathematically described as a gauss-diffusion process. The first-passage-time probability density, miming in such a context the firing probability density, is evaluated by either the Volterra integral equation of Buonocore, Nobile, and Ricciardi (1987) or, when possible, by the asymptotics of Giorno, Nobile, and Ricciardi (1990). The model examined here represents an extension of the classic leaky integrate-and-fire one based on the Ornstein-Uhlenbeck process in that it is in principle compatible with the inclusion of some other physiological characteristics such as relative refractoriness. It also allows finer tuning possibilities in view of its accounting for certain qualitative as well as quantitative features, such as the behavior of the time course of the membrane potential prior to firings and the computation of experimentally measurable statistical descriptors of the firing time: mean, median, coefficient of variation, and skewness. Finally, implementations of this model are provided in connection with certain experimental evidence discussed in the literature.

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79. Neuronal Population Decoding Explains the Change in Signal Detection Sensitivity Caused by Task-Irrelevant Perceptual Bias

Satohiro Tajima, Hiromasa Takemura, Ikuya Murakami, Masato Okada

https://doi.org/10.1162/neco_a_00019 | pp. 2586-2614

Spatiotemporal context in sensory stimulus has profound effects on neural responses and perception, and it sometimes affects task difficulty. Recently reported experimental data suggest that human detection sensitivity to motion in a target stimulus can be enhanced by adding a slow surrounding motion in an orthogonal direction, even though the illusory motion component caused by the surround is not relevant to the task. It is not computationally clear how the task-irrelevant component of motion modulates the subject's sensitivity to motion detection. In this study, we investigated the effects of encoding biases on detection performance by modeling the stochastic neural population activities. We modeled two types of modulation on the population activity profiles caused by a contextual stimulus: one type is identical to the activity evoked by a physical change in the stimulus, and the other is expressed more simply in terms of response gain modulation. For both encoding schemes, the motion detection performance of the ideal observer is enhanced by a task-irrelevant, additive motion component, replicating the properties observed for real subjects. The success of these models suggests that human detection sensitivity can be characterized by a noisy neural encoding that limits the resolution of information transmission in the cortical visual processing

pathway. On the other hand, analyses of the neuronal contributions to the task predict that the effective cell populations differ between the two encoding schemes, posing a question concerning the decoding schemes that the nervous system used during illusory states.

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80. Spiking Neural P Systems with Weights

Jun Wang, Hendrik Jan Hoogeboom, Linqiang Pan, Gheorghe P■un, Mario J. Pérez-Jímenez
https://doi.org/10.1162/neco_a_00022 | pp. 2615-2646

A variant of spiking neural P systems with positive or negative weights on synapses is introduced, where the rules of a neuron fire when the potential of that neuron equals a given value. The involved values-weights, firing thresholds, potential consumed by each rule-can be real (computable) numbers, rational numbers, integers, and natural numbers. The power of the obtained systems is investigated. For instance, it is proved that integers (very restricted: 1, -1 for weights, 1 and 2 for firing thresholds, and as parameters in the rules) suffice for computing all Turing computable sets of numbers in both the generative and the accepting modes. When only natural numbers are used, a characterization of the family of semilinear sets of numbers is obtained. It is shown that spiking neural P systems with weights can efficiently solve computationally hard problems in a nondeterministic way. Some open problems and suggestions for further research are formulated.

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81. A New Bidimensional Neural Field Model with Heterogeneous Connection Topology

Mouhamad Jradeh
https://doi.org/10.1162/neco_a_00017 | pp. 2647-2654

This letter develops a new mathematical cerebral activity model for the synaptic value of neural population.

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82. Convergence Analysis of Three Classes of Split-Complex Gradient Algorithms for Complex-Valued Recurrent Neural Networks

Dongpo Xu, Huisheng Zhang, Lijun Liu
https://doi.org/10.1162/neco_a_00021 | pp. 2655-2677

This letter presents a unified convergence analysis of the split-complex nonlinear gradient descent (SCNGD) learning algorithms for complex-valued recurrent neural networks, covering three classes of SCNGD algorithms: standard SCNGD, normalized SCNGD, and adaptive normalized SCNGD. We prove that if the activation functions are of split-complex type and some conditions are satisfied, the error function is monotonically decreasing during the training iteration process, and the gradients of the error function with respect to the real and imaginary parts of the weights converge to zero. A strong convergence result is also obtained under the assumption that the error function has only a finite number of stationary points. The simulation results are given to support the theoretical analysis.

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83. Large-Margin Classification in Infinite Neural Networks

Youngmin Cho, Lawrence K. Saul

https://doi.org/10.1162/neco_a_00018 | pp. 2678-2697

We introduce a new family of positive-definite kernels for large margin classification in support vector machines (SVMs). These kernels mimic the computation in large neural networks with one layer of hidden units. We also show how to derive new kernels, by recursive composition, that may be viewed as mapping their inputs through a series of nonlinear feature spaces. These recursively derived kernels mimic the computation in deep networks with multiple hidden layers. We evaluate SVMs with these kernels on problems designed to illustrate the advantages of deep architectures. Compared to previous benchmarks, we find that on some problems, these SVMs yield state-of-the-art results, beating not only other SVMs but also deep belief nets.

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84. The MEE Principle in Data Classification: A Perceptron-Based Analysis

Luís M. Silva, Joaquim Marques de Sá, Lu■■s A. Alexandre

https://doi.org/10.1162/neco_a_00013 | pp. 2698-2728

This letter focuses on the issue of whether risk functionals derived from information-theoretic principles, such as Shannon or Rényi's entropies, are able to cope with the data classification problem in both the sense of attaining the risk functional minimum and implying the minimum probability of error allowed by the family of functions implemented by the classifier, here denoted by min Pe. The analysis of this so-called minimization of error entropy (MEE) principle is carried out in a single perceptron with continuous activation functions, yielding continuous error distributions. In spite of the fact that the analysis is restricted to single perceptrons, it reveals a large spectrum of behaviors that MEE can be expected to exhibit in both theory and practice. In what concerns the theoretical MEE, our study clarifies the role of the parameters controlling the perceptron activation function (of the squashing type) in often reaching the minimum probability of error. Our study also clarifies the role of the kernel density estimator of the error density in achieving the minimum probability of error in practice.

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85. Comparing Classification Methods for Longitudinal fMRI Studies

Tanya Schmah, Grigori Yourganov, Richard S. Zemel, Geoffrey E. Hinton, Steven L. Small, Stephen C. Strother

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We compare 10 methods of classifying fMRI volumes by applying them to data from a longitudinal study of stroke recovery: adaptive Fisher's linear and quadratic discriminant; gaussian naive Bayes; support vector machines with linear, quadratic, and radial basis function (RBF) kernels; logistic regression; two novel methods based on pairs of restricted Boltzmann machines (RBM); and K-nearest neighbors. All methods were tested on three binary classification tasks, and their out-of-sample classification accuracies are compared. The relative performance of the methods varies considerably across subjects and classification tasks. The best overall performers were adaptive quadratic discriminant, support vector machines with RBF kernels, and generatively trained pairs of RBMs.

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86. How Are Lateral Chromatic Interactions Computed from Cone Signals?

C Wehrhahn, Harald Teufel

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A small gray test field superimposed on a large colored background appears tinted in a color complementary to that of the surround. We tested the hypothesis whether photoreceptor sensitivity in the test field is altered in the presence of a colored surround. We investigated this effect using dichoptic viewing conditions. With the left eye, subjects viewed a small gray target superimposed on a large colored background. The right eye saw a gray target superimposed on a large gray background. When the two images were fused, observers perceived one homogeneous background and two targets. Observers matched the color of the target seen by the right eye to that seen by the left eye. A modified two-stage model for chromatic induction assuming that both receptor and postreceptor mechanisms contribute to the shift in color was fitted to the matched settings. We find that the dichoptic viewing effects presented here are well explained by an approximately equal contribution of receptor and postreceptor processes to the perceived shift in color.

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87. A Fast ■_p Spike Alignment Metric

Alexander Dubbs, Brad A. Seiler, Marcelo O. Magnasco

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The metrization of the space of neural responses is an ongoing research program seeking to find natural ways to describe, in geometrical terms, the sets of possible activities in the brain. One component of this program is spike metrics-notions of distance between two spike trains recorded from a neuron. Alignment spike metrics work by identifying "equivalent" spikes in both trains. We present an alignment spike metric having L(p) underlying geometrical structure; the L(2) version is Euclidean and is suitable for further embedding in Euclidean spaces by multidimensional scaling methods or related procedures. We show how to implement a fast algorithm for the computation of this metric based on bipartite graph matching theory.

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88. The Neuronal Replicator Hypothesis

Chrisantha Fernando, Richard A. Goldstein, Eörs Szathmáry

https://doi.org/10.1162/neco_a_00031 | pp. 2809-2857

We propose that replication (with mutation) of patterns of neuronal activity can occur within the brain using known neurophysiological processes. Thereby evolutionary algorithms implemented by neuro- nal circuits can play a role in cognition. Replication of structured neuronal representations is assumed in several cognitive architectures. Replicators overcome some limitations of selectionist models of neuronal search. Hebbian learning is combined with replication to structure exploration on the basis of associations learned in the past. Neuromodulatory gating of sets of bistable neurons allows patterns of activation to be copied with mutation. If the probability of copying a set is related to the utility of that set, then an evolutionary algorithm can be implemented at rapid timescales in the brain. Populations of neuronal replicators can undertake a more rapid and stable search than can be achieved by serial modification of a single solution. Hebbian learning added to neuronal replication allows a powerful structuring of variability capable of learning the location of a global optimum from multiple previously visited local optima. Replication of solutions can solve the problem of catastrophic forgetting in the stability-plasticity dilemma. In short, neuronal replication is

essential to explain several features of flexible cognition. Predictions are made for the experimental validation of the neuronal replicator hypothesis.

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89. Rademacher Chaos Complexities for Learning the Kernel Problem

Yiming Ying, Colin Campbell

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We develop a novel generalization bound for learning the kernel problem. First, we show that the generalization analysis of the kernel learning problem reduces to investigation of the suprema of the Rademacher chaos process of order 2 over candidate kernels, which we refer to as Rademacher chaos complexity. Next, we show how to estimate the empirical Rademacher chaos complexity by well-established metric entropy integrals and pseudo-dimension of the set of candidate kernels. Our new methodology mainly depends on the principal theory of U-processes and entropy integrals. Finally, we establish satisfactory excess generalization bounds and misclassification error rates for learning gaussian kernels and general radial basis kernels.

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90. A Conditional Entropy Minimization Criterion for Dimensionality Reduction and Multiple Kernel Learning

Hideitsu Hino, Noboru Murata

https://doi.org/10.1162/neco_a_00027 | pp. 2887-2923

Reducing the dimensionality of high-dimensional data without losing its essential information is an important task in information processing. When class labels of training data are available, Fisher discriminant analysis (FDA) has been widely used. However, the optimality of FDA is guaranteed only in a very restricted ideal circumstance, and it is often observed that FDA does not provide a good classification surface for many real problems. This letter treats the problem of supervised dimensionality reduction from the viewpoint of information theory and proposes a framework of dimensionality reduction based on class-conditional entropy minimization. The proposed linear dimensionality-reduction technique is validated both theoretically and experimentally. Then, through kernel Fisher discriminant analysis (KFDA), the multiple kernel learning problem is treated in the proposed framework, and a novel algorithm, which iteratively optimizes the parameters of the classification function and kernel combination coefficients, is proposed. The algorithm is experimentally shown to be comparable to or outperforms KFDA for large-scale benchmark data sets, and comparable to other multiple kernel learning techniques on the yeast protein function annotation task.

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91. Window-Based Example Selection in Learning Vector Quantization

Aree Witoelar, A. Ghosh, J. J. G. de Vries, Barbara Hammer, Michael Biehl

https://doi.org/10.1162/neco_a_00030 | pp. 2924-2961

A variety of modifications have been employed to learning vector quantization (LVQ) algorithms using either crisp or soft windows for selection of data. Although these schemes have been shown in practice to improve performance, a theoretical study on the influence of windows has so far been limited. Here we rigorously analyze the influence of windows in a controlled environment of gaussian mixtures in high dimensions.

Concepts from statistical physics and the theory of online learning allow an exact description of the training dynamics, yielding typical learning curves, convergence properties, and achievable generalization abilities. We compare the performance and demonstrate the advantages of various algorithms, including LVQ 2.1, generalized LVQ (GLVQ), Learning from Mistakes (LFM) and Robust Soft LVQ (RSLVQ). We find that the selection of the window parameter highly influences the learning curves but not, surprisingly, the asymptotic performances of LVQ 2.1 and RSLVQ. Although the prototypes of LVQ 2.1 exhibit divergent behavior, the resulting decision boundary coincides with the optimal decision boundary, thus yielding optimal generalization ability.

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92. A Novel Recurrent Neural Network with Finite-Time Convergence for Linear Programming

Qingshan Liu, Jinde Cao, Guanrong Chen

https://doi.org/10.1162/neco_a_00029 | pp. 2962-2978

In this letter, a novel recurrent neural network based on the gradient method is proposed for solving linear programming problems. Finite-time convergence of the proposed neural network is proved by using the Lyapunov method. Compared with the existing neural networks for linear programming, the proposed neural network is globally convergent to exact optimal solutions in finite time, which is remarkable and rare in the literature of neural networks for optimization. Some numerical examples are given to show the effectiveness and excellent performance of the new recurrent neural network.

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93. A Theoretical Basis for Emergent Pattern Discrimination in Neural Systems Through Slow Feature Extraction

Stefan Klampfl, Wolfgang Maass

https://doi.org/10.1162/neco_a_00050 | pp. 2979-3035

Neurons in the brain are able to detect and discriminate salient spatiotemporal patterns in the firing activity of presynaptic neurons. It is open how they can learn to achieve this, especially without the help of a supervisor. We show that a well-known unsupervised learning algorithm for linear neurons, slow feature analysis (SFA), is able to acquire the discrimination capability of one of the best algorithms for supervised linear discrimination learning, the Fisher linear discriminant (FLD), given suitable input statistics. We demonstrate the power of this principle by showing that it enables readout neurons from simulated cortical microcircuits to learn without any supervision to discriminate between spoken digits and to detect repeated firing patterns that are embedded into a stream of noise spike trains with the same firing statistics. Both these computer simulations and our theoretical analysis show that slow feature extraction enables neurons to extract and collect information that is spread out over a trajectory of firing states that lasts several hundred ms. In addition, it enables neurons to learn without supervision to keep track of time (relative to a stimulus onset, or the initiation of a motor response). Hence, these results elucidate how the brain could compute with trajectories of firing states rather than only with fixed point attractors. It also provides a theoretical basis for understanding recent experimental results on the emergence of view- and position-invariant classification of visual objects in inferior temporal cortex.

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94. Change-Based Inference in Attractor Nets: Linear Analysis

Reza Moazzezi, Peter Dayan

https://doi.org/10.1162/neco_a_00051 | pp. 3036-3061

One standard interpretation of networks of cortical neurons is that they form dynamical attractors. Computations such as stimulus estimation are performed by mapping inputs to points on the networks' attractive manifolds. These points represent population codes for the stimulus values. However, this standard interpretation is hard to reconcile with the observation that the firing rates of such neurons constantly change following presentation of stimuli. We have recently suggested an alternative interpretation according to which computations are realized by systematic changes in the states of such networks over time. This way of performing computations is fast, accurate, readily learnable, and robust to various forms of noise. Here we analyze the computation of stimulus discrimination in this change-based setting, relating it directly to the computation of stimulus estimation in the conventional attractor-based view. We use a common linear approximation to compare the two methods and show that perfect performance at estimation implies chance performance at discrimination.

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95. Internal-Time Temporal Difference Model for Neural Value-Based Decision Making

Hiroyuki Nakahara, Sivaramakrishnan Kaveri

https://doi.org/10.1162/neco_a_00049 | pp. 3062-3106

The temporal difference (TD) learning framework is a major paradigm for understanding value-based decision making and related neural activities (e.g., dopamine activity). The representation of time in neural processes modeled by a TD framework, however, is poorly understood. To address this issue, we propose a TD formulation that separates the time of the operator (neural valuation processes), which we refer to as internal time, from the time of the observer (experiment), which we refer to as conventional time. We provide the formulation and theoretical characteristics of this TD model based on internal time, called internal-time TD, and explore the possible consequences of the use of this model in neural value-based decision making. Due to the separation of the two times, internal-time TD computations, such as TD error, are expressed differently, depending on both the time frame and time unit. We examine this operator-observer problem in relation to the time representation used in previous TD models. An internal time TD value function exhibits the co-appearance of exponential and hyperbolic discounting at different delays in intertemporal choice tasks. We further examine the effects of internal time noise on TD error, the dynamic construction of internal time, and the modulation of internal time with the internal time hypothesis of serotonin function. We also relate the internal TD formulation to research on interval timing and subjective time.

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96. Phase Coupling Estimation from Multivariate Phase Statistics

Charles F. Cadieu, Kilian Koepsell

https://doi.org/10.1162/neco_a_00048 | pp. 3107-3126

Coupled oscillators are prevalent throughout the physical world. Dynamical system formulations of weakly coupled oscillator systems have proven effective at capturing the properties of real-world systems and are compelling models of neural systems. However, these formulations usually deal with the forward problem:

simulating a system from known coupling parameters. Here we provide a solution to the inverse problem: determining the coupling parameters from measurements. Starting from the dynamic equations of a system of symmetrically coupled phase oscillators, given by a nonlinear Langevin equation, we derive the corresponding equilibrium distribution. This formulation leads us to the maximum entropy distribution that captures pairwise phase relationships. To solve the inverse problem for this distribution, we derive a closed-form solution for estimating the phase coupling parameters from observed phase statistics. Through simulations, we show that the algorithm performs well in high dimensions (d = 100) and in cases with limited data (as few as 100 samples per dimension). In addition, we derive a regularized solution to the estimation and show that the resulting procedure improves performance when only a limited amount of data is available. Because the distribution serves as the unique maximum entropy solution for pairwise phase statistics, phase coupling estimation can be broadly applied in any situation where phase measurements are made. Under the physical interpretation, the model may be used for inferring coupling relationships within cortical networks.

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97. Neural Decoding with Hierarchical Generative Models

Marcel van Gerven, Floris P. de Lange, Tom Heskes

https://doi.org/10.1162/neco_a_00047 | pp. 3127-3142

Recent research has shown that reconstruction of perceived images based on hemodynamic response as measured with functional magnetic resonance imaging (fMRI) is starting to become feasible. In this letter, we explore reconstruction based on a learned hierarchy of features by employing a hierarchical generative model that consists of conditional restricted Boltzmann machines. In an unsupervised phase, we learn a hierarchy of features from data, and in a supervised phase, we learn how brain activity predicts the states of those features. Reconstruction is achieved by sampling from the model, conditioned on brain activity. We show that by using the hierarchical generative model, we can obtain good-quality reconstructions of visual images of handwritten digits presented during an fMRI scanning session.

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98. Robust Observer-Based Tracking Control of Hodgkin-Huxley Neuron Systems Under Environmental Disturbances

Bor■Sen Chen, Chengwei Li

https://doi.org/10.1162/neco_a_00053 | pp. 3143-3178

A nervous system consists of a large number of highly interconnected nerve cells. Nerve cells communicate by generation and transmission of short electrical pulses (action potential). In addition, membrane voltage is the only measurable state in nervous systems. A robust observer-based model reference tracking control is proposed for Hodgkin-Huxley (HH) neuron systems to generate a desired reference response in spite of environmental noises, uncertain initial values, and diffusion currents from other interconnected nerve cells. In order to simplify the robust tracking control design of nonlinear stochastic HH neuron systems, a fuzzy interpolation method is employed to interpolate several linear stochastic systems to approximate a nonlinear stochastic HH neuron system so that the nonlinear robust tracking control problem can be solved by the linear matrix inequality (LMI) technique with the help of Robust Control Toolbox in Matlab. The proposed robust observer-based tracking control scheme can provide new methods for desired action potential generation, suppression of oscillations, and blockage of action potential transmission under environmental

noise and diffusion currents. These new methods are useful for patients with different neuron system dysfunctions. Finally, three simulation examples of tracking control of nervous systems are given to illustrate the design procedure and confirm the tracking performance of the proposed method.

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99. Psychophysically Tuned Divisive Normalization Approximately Factorizes the PDF of Natural Images

Jesús Malo, Valero Laparra

https://doi.org/10.1162/neco_a_00046 | pp. 3179-3206

The conventional approach in computational neuroscience in favor of the efficient coding hypothesis goes from image statistics to perception. It has been argued that the behavior of the early stages of biological visual processing (e.g., spatial frequency analyzers and their nonlinearities) may be obtained from image samples and the efficient coding hypothesis using no psychophysical or physiological information. In this work we address the same issue in the opposite direction: from perception to image statistics. We show that psychophysically fitted image representation in V1 has appealing statistical properties, for example, approximate PDF factorization and substantial mutual information reduction, even though no statistical information is used to fit the V1 model. These results are complementary evidence in favor of the efficient coding hypothesis.

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100. Deep, Big, Simple Neural Nets for Handwritten Digit Recognition

Dan Claudiu Cire■an, Ueli Meier, Luca Maria Gambardella, Jürgen Schmidhuber

https://doi.org/10.1162/neco_a_00052 | pp. 3207-3220

Good old online backpropagation for plain multilayer perceptrons yields a very low 0.35% error rate on the MNIST handwritten digits benchmark. All we need to achieve this best result so far are many hidden layers, many neurons per layer, numerous deformed training images to avoid overfitting, and graphics cards to greatly speed up learning.

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101. Least Square Regression with l^p-Coefficient Regularization

Hongzhi Tong, Di■Rong Chen, Fenghong Yang

https://doi.org/10.1162/neco_a_00044 | pp. 3221-3235

The selection of the penalty functional is critical for the performance of a regularized learning algorithm, and thus it deserves special attention. In this article, we present a least square regression algorithm based on l p -coefficient regularization. Comparing with the classical regularized least square regression, the new algorithm is different in the regularization term. Our primary focus is on the error analysis of the algorithm. An explicit learning rate is derived under some ordinary assumptions.

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102. Volume 22 Index by Author

(authors unknown)
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